Maximal and Closed Frequent Itemsets

Introduction

- In previous, we talked about **frequent itemset mining.**
- It is a data analysis task that has many applications.
- However, a problem is that the number of frequent itemsets is often very large.
- Besides, frequent itemsets often have some form of redundancy.
- Today, we will talk about this problem and a solution, which is to discover **concise representations of frequent itemsets.**

Frequent itemset mining

Input: A transaction database

Transaction	Items appearing in the transaction
T1	{pasta, lemon, bread, orange}
T2	{pasta, lemon}
T3	{pasta, orange, cake}
T4	{pasta, lemon, orange cake}

minsup = 2

Frequent itemset mining

Input: A transaction database

Transaction	Items appearing in the transaction
T1	{pasta, lemon, bread, orange}
T2	{pasta, lemon}
T3	{pasta, orange, cake}
T4	{pasta, lemon, orange cake}

minsup = 2

Output: The frequent itemsets
{lemon}, {pasta}, {orange}, {cake}, {lemon, pasta}, {lemon, orange}, {pasta, orange}, {pasta, cake}, {orange, cake}, {lemon, pasta, orange}, {pasta, orange, cake}

Frequent itemsets

{pasta}	support = 4
{lemon}	support = 3
{orange}	support = 3
{cake}	support = 2
{pasta, lemon}	support: 3
{pasta, orange}	support: 3
{pasta, cake}	support: 2
{lemon, orange}	support: 2
{orange, cake}	support: 2
{pasta, lemon, orange}	support: 2
{pasta, orange, cake}	support: 2

Frequent itemsets

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{pasta, lemon}	support: 3
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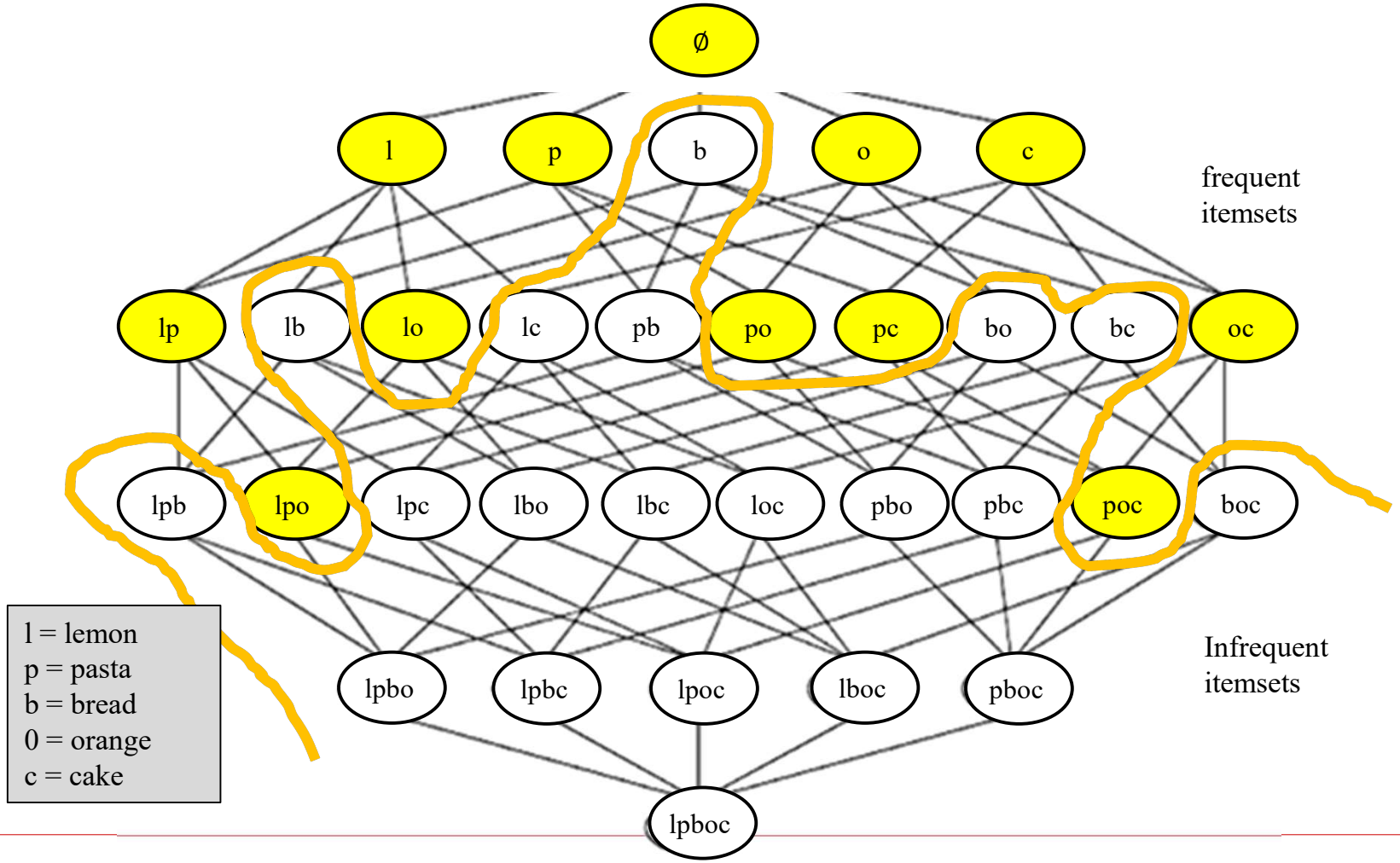
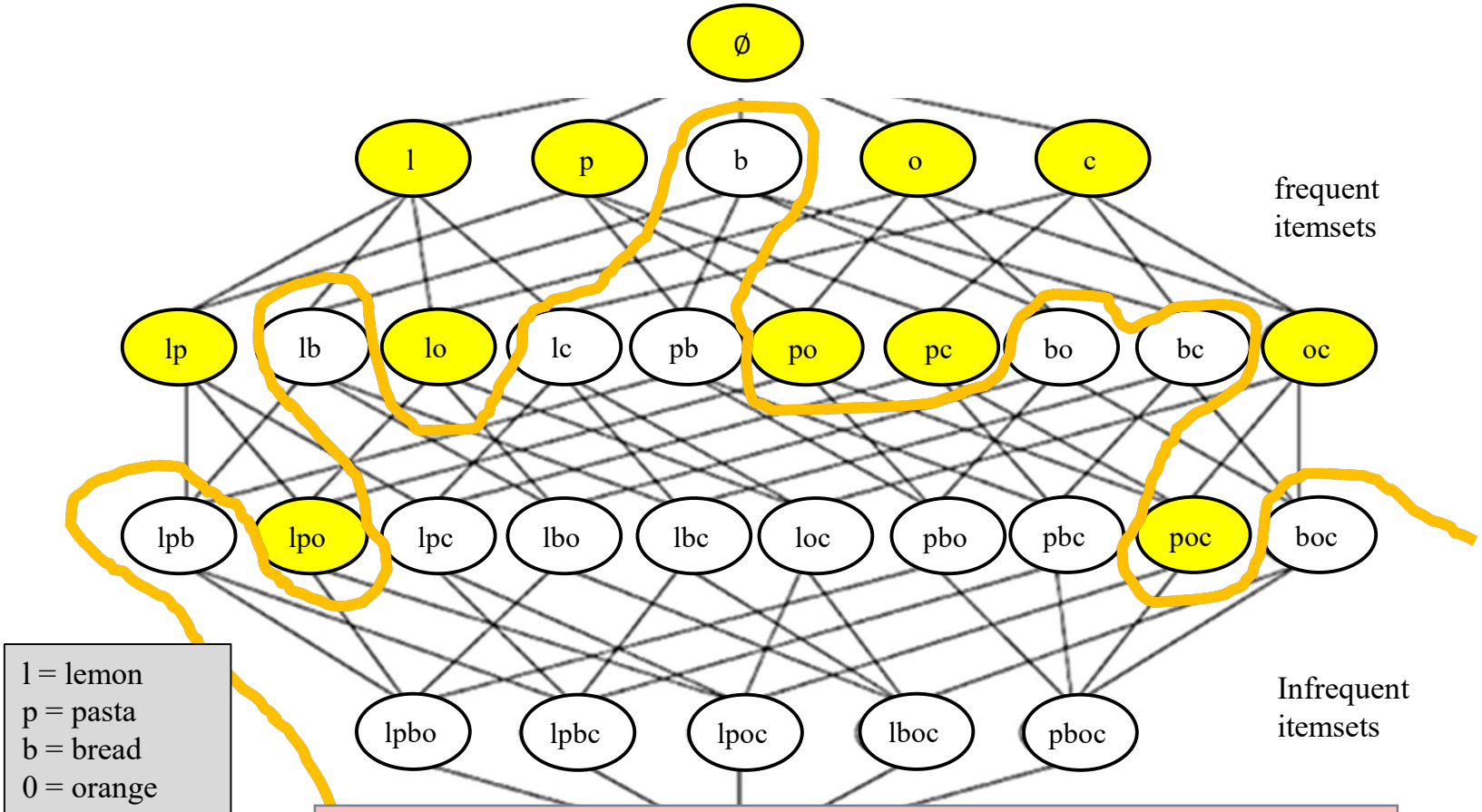
{pasta, cake}	support: 2
{lemon, orange}	support: 2
{orange, cake}	support: 2
{pasta, lemon, orange}	support: 2
{pasta, orange, cake}	support: 2

There are 11 frequent itemsets!

minsup =2 The frequent itemsets

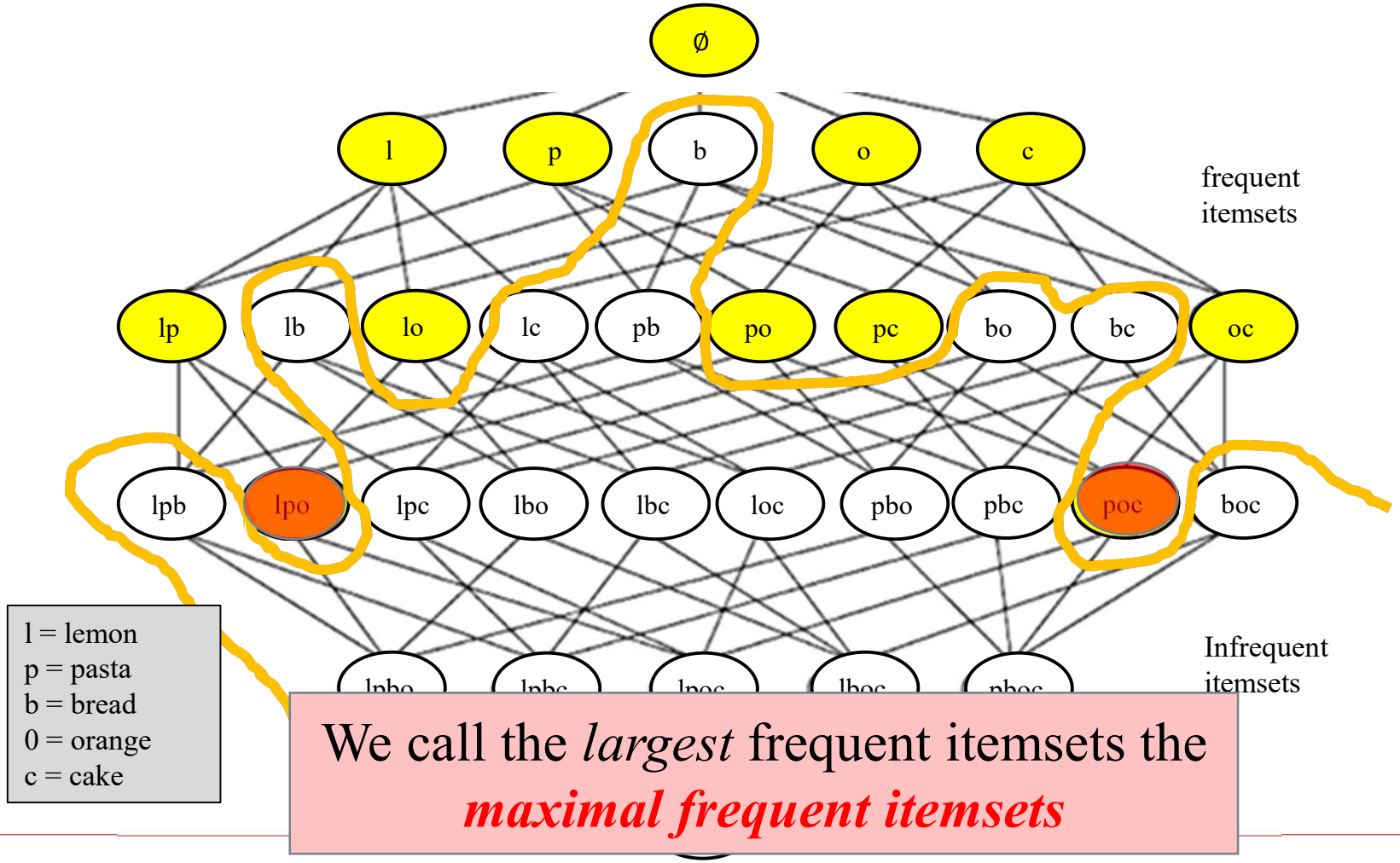


minsup =2 The frequent itemsets



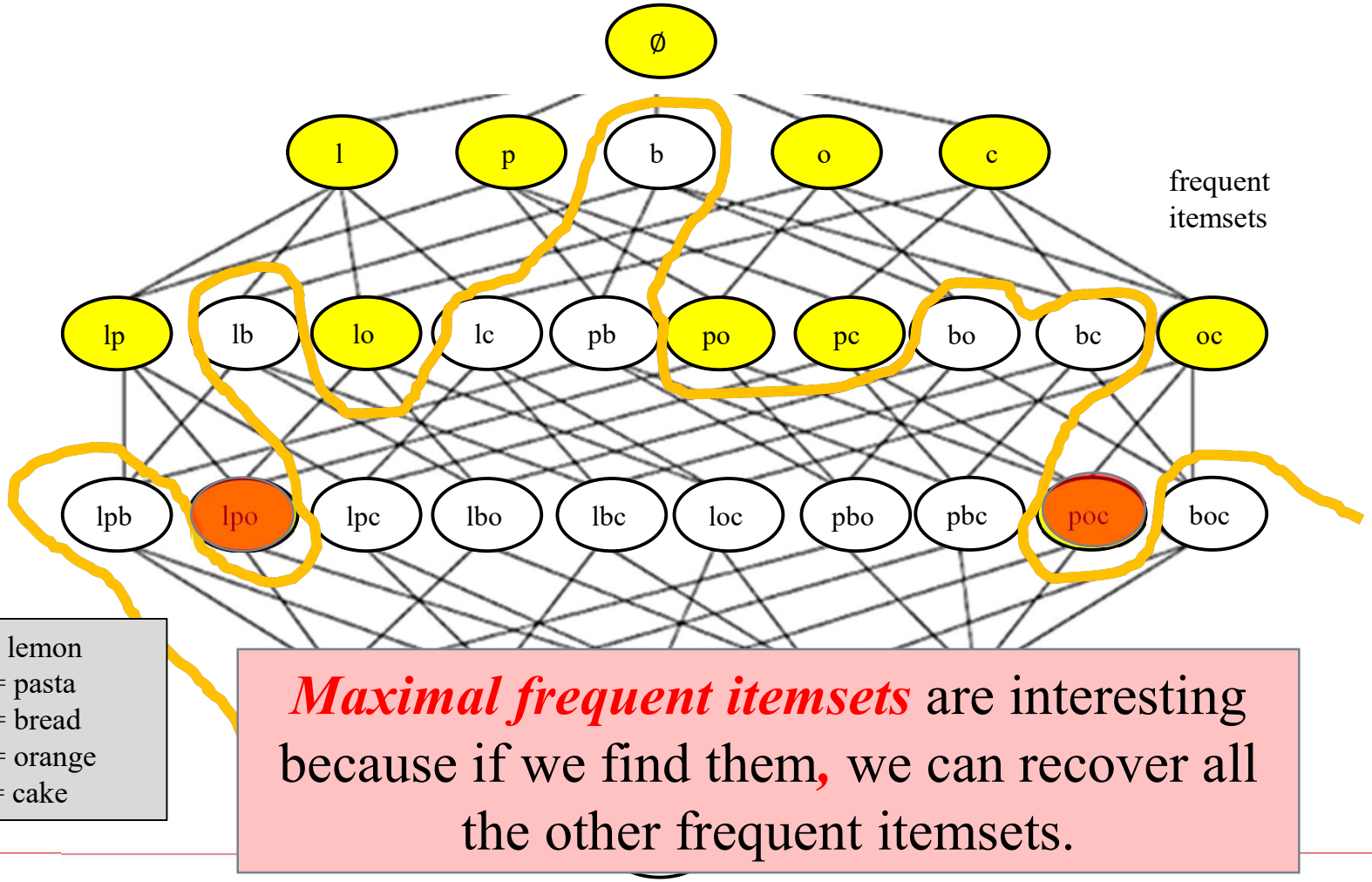
We can observe that if an itemset is frequent then all its subsets are frequent

minsup =2 The frequent itemsets



We call the *largest* frequent itemsets the ***maximal frequent itemsets***

minsup =2 The frequent itemsets



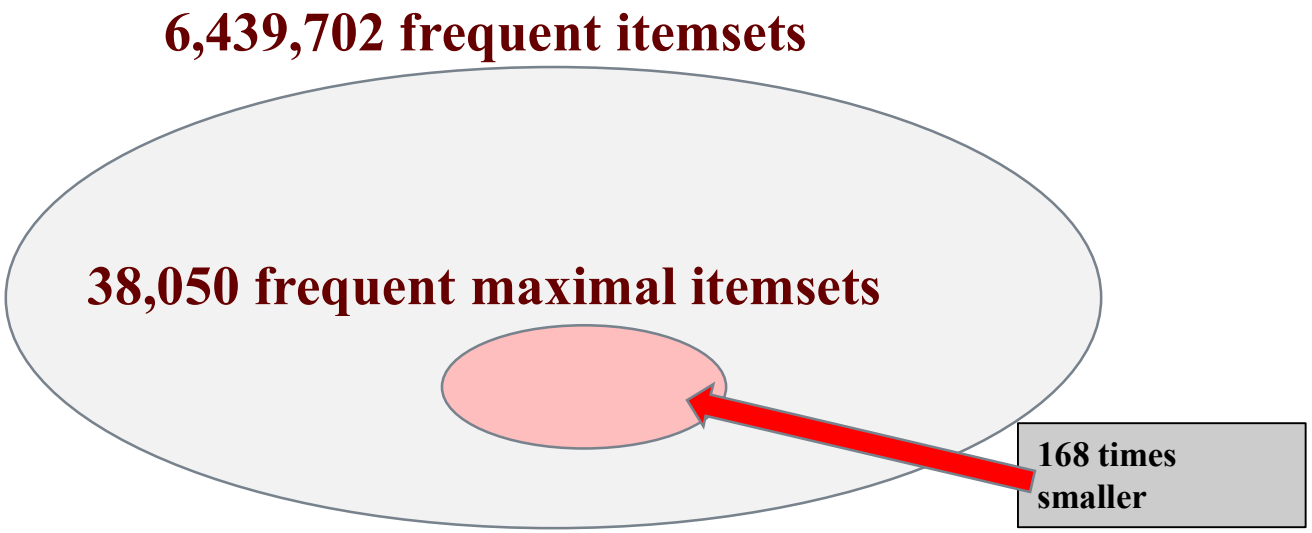
l = lemon
p = pasta
b = bread
0 = orange
c = cake

Maximal frequent itemsets are interesting because if we find them, we can recover all the other frequent itemsets.

Maximal itemsets are compact

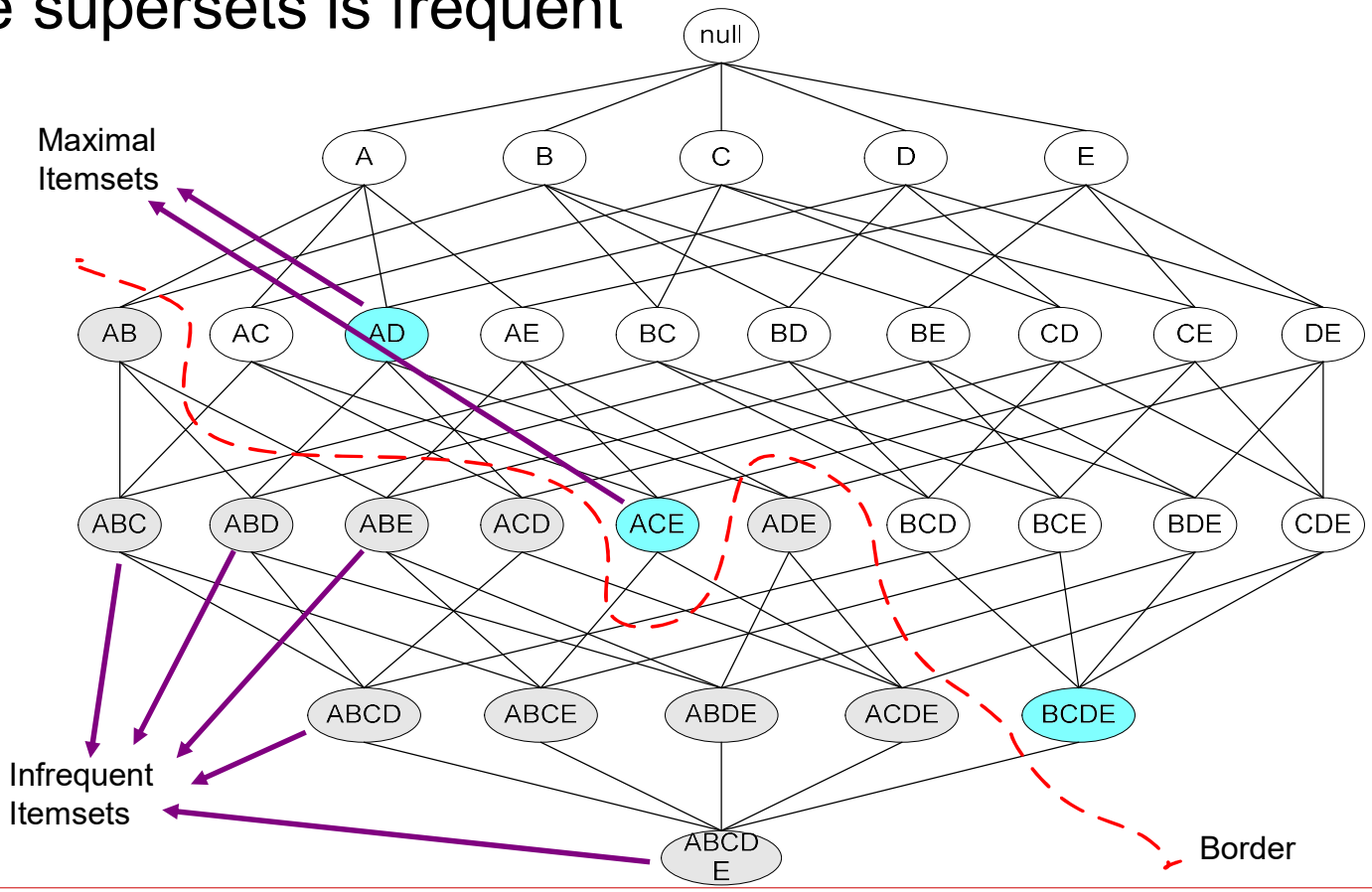
If we find the maximal frequent itemsets, we can find **much less** itemsets.

Example: ***Chess*** dataset and minsup =0.4



Maximal Frequent Itemset

An itemset X is maximal frequent if it is frequent and none of its immediate supersets is frequent



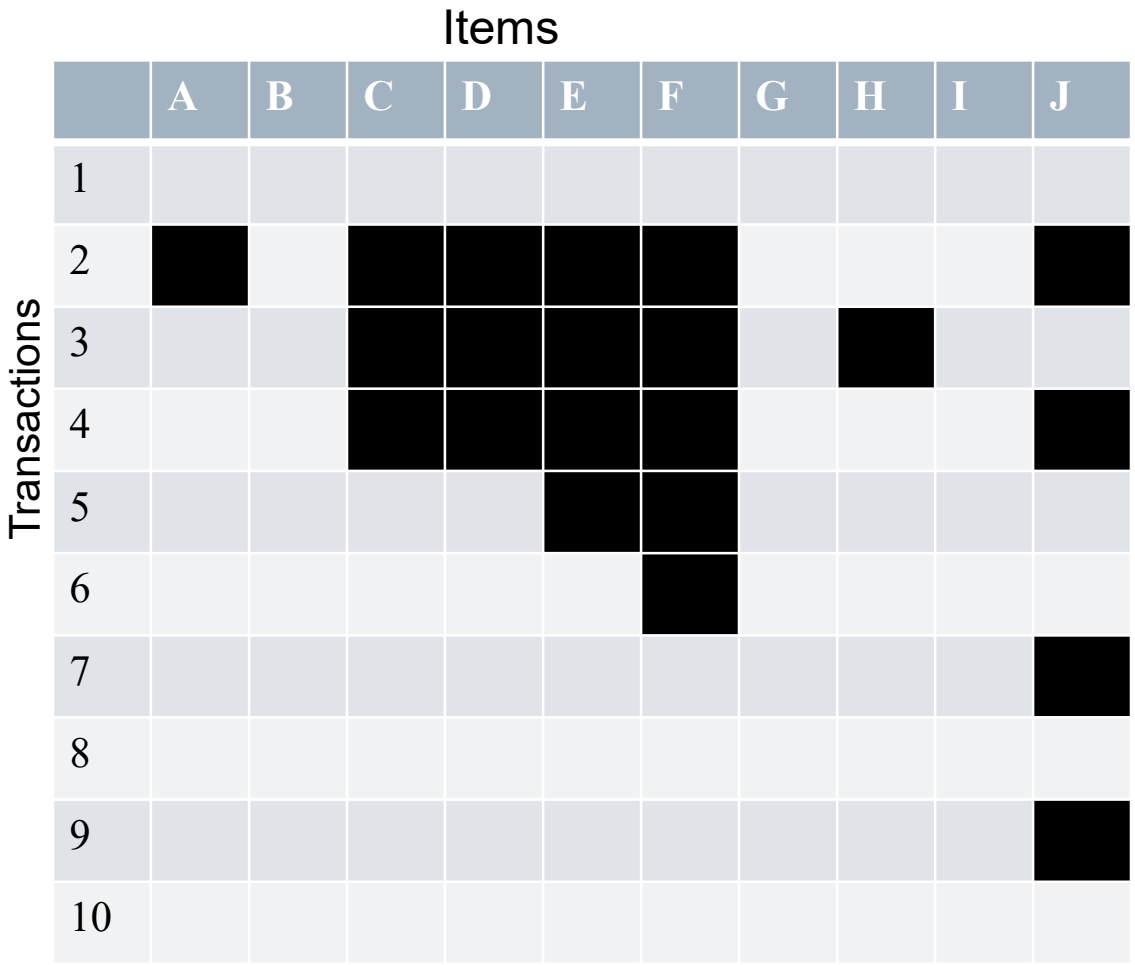
An illustrative example

Support threshold (by count) : 5

Frequent itemsets: ?

Maximal itemsets: ?

An illustrative example



Support threshold (by count) : 5

Frequent itemsets: {F}

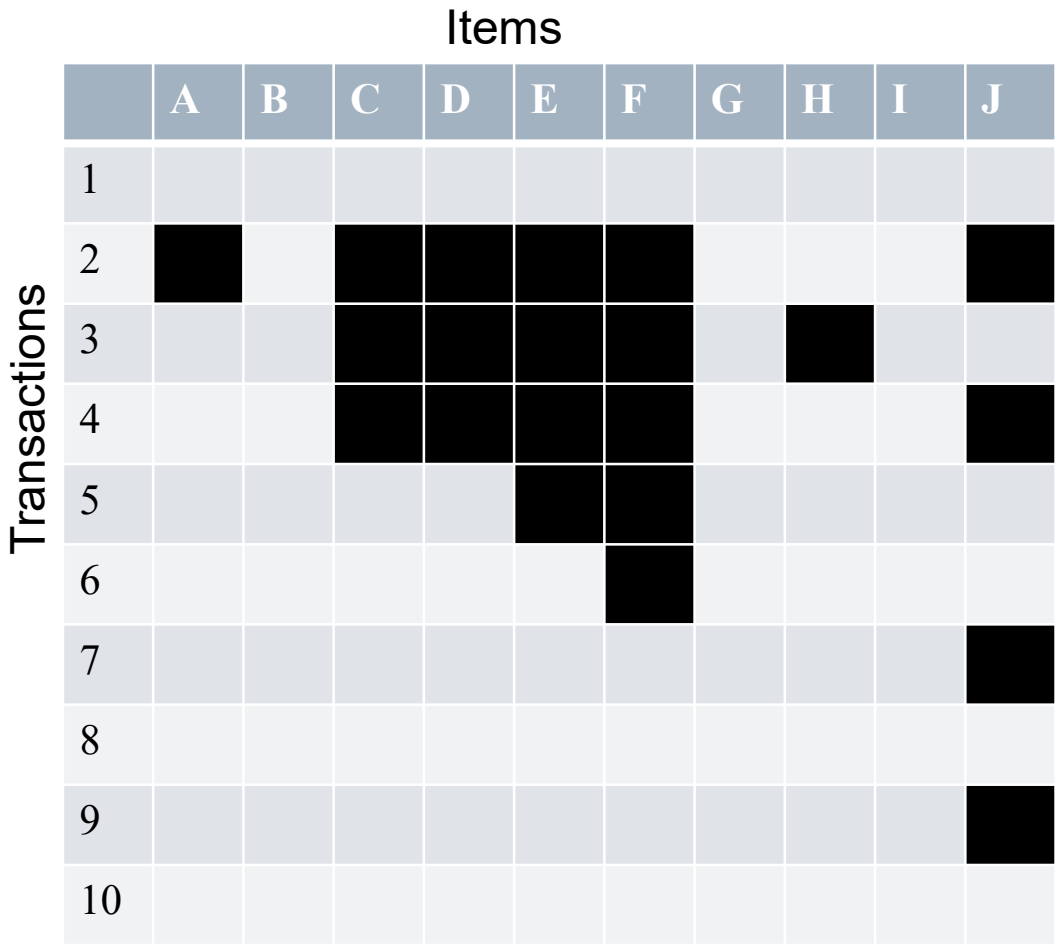
Maximal itemsets: {F}

Support threshold (by count): 4

Frequent itemsets: ?

Maximal itemsets: ?

An illustrative example

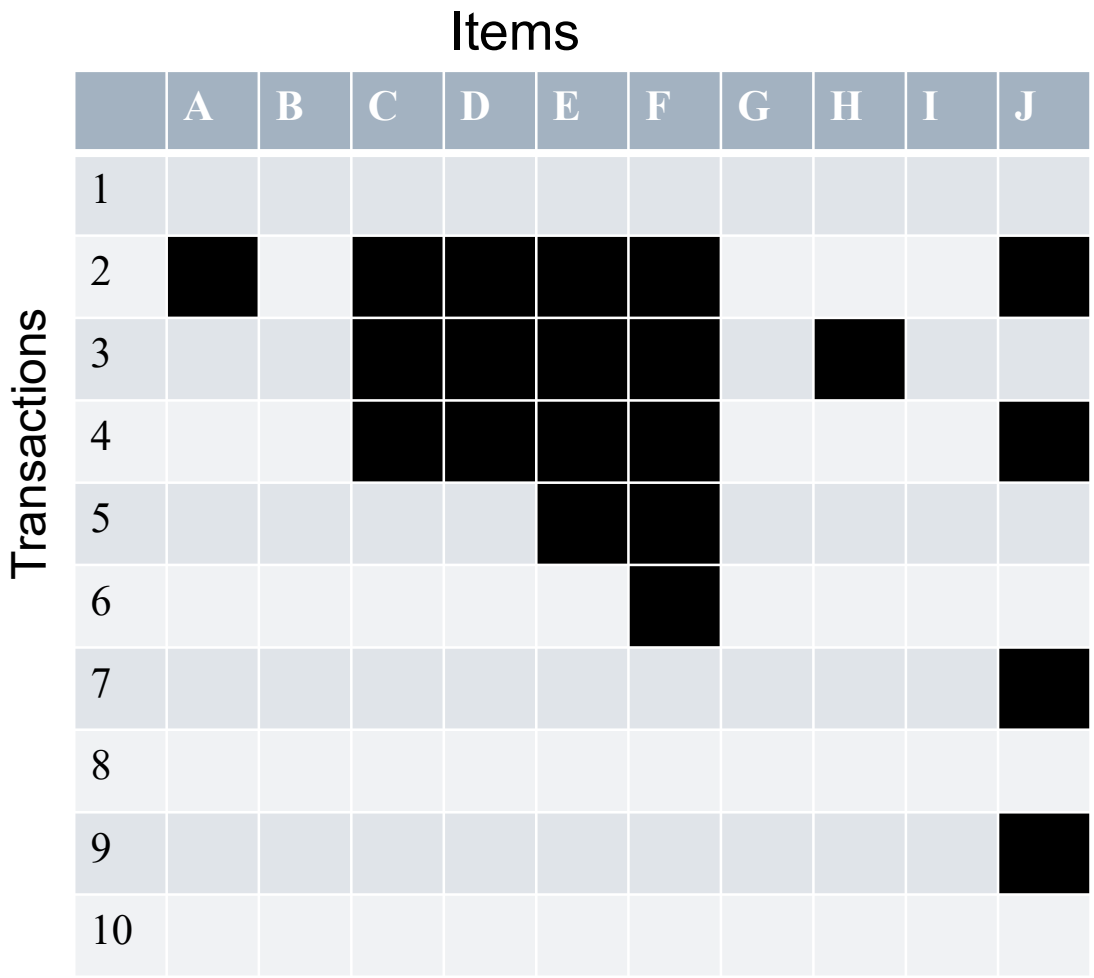


Support threshold (by count) : 5
Frequent itemsets: {F}
Maximal itemsets: {F}

Support threshold (by count): 4
Frequent itemsets: {E}, {F}, {E,F}, {J}
Maximal itemsets: {E,F}, {J}

Support threshold (by count): 3
Frequent itemsets: ?
Maximal itemsets: ?

An illustrative example



Support threshold (by count) : 5

Frequent itemsets: {F}

Maximal itemsets: {F}

Support threshold (by count): 4

Frequent itemsets: {E}, {F}, {E,F}, {J}

Maximal itemsets: {E,F}, {J}

Support threshold (by count): 3

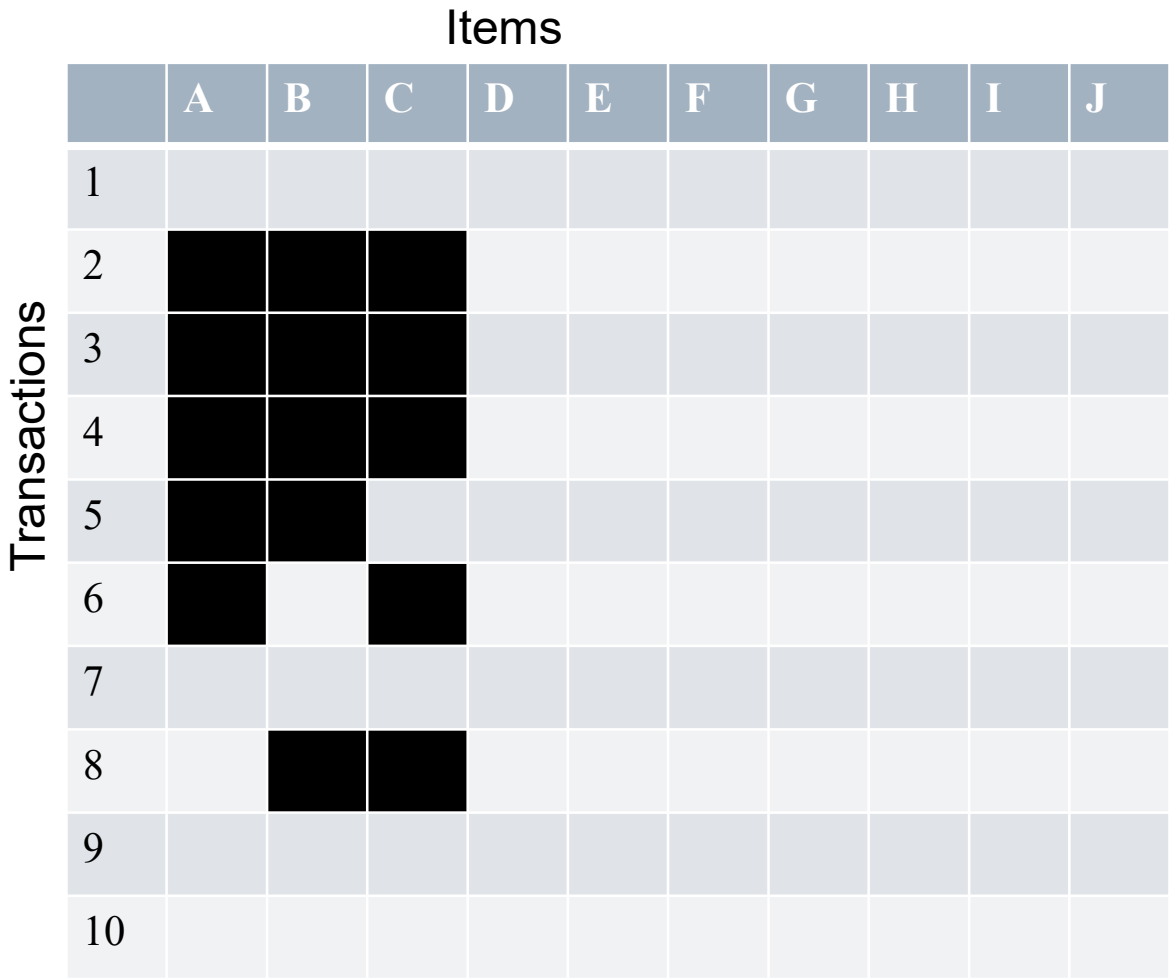
Frequent itemsets:

All subsets of {C,D,E,F} + {J}

Maximal itemsets:

{C,D,E,F}, {J}

Another illustrative example



Support threshold (by count) : 5

Maximal itemsets: {A}, {B}, {C}

Support threshold (by count): 4

Maximal itemsets: {A,B}, {A,C},{B,C}

Support threshold (by count): 3

Maximal itemsets: {A,B,C}

How to find the maximal itemsets?

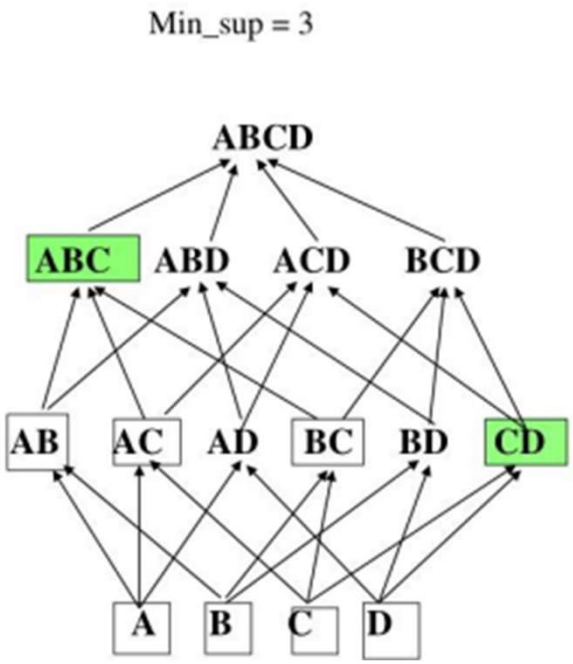
- **Naïve approach**
  - by post-processing
  - **inefficient!**
- **Efficient algorithms:**
  - GenMax: a modified version of ECLAT
  - FPMax: a modified version of FP-Growth
  - LCMMax: a modified version of LCM
  - … and many others!

- Those algorithms adopt various strategies to find the maximal itemsets without having to find all frequent itemsets.

GenMax algorithm

- GenMax is an algorithm to find the exact MFI
- From: "Efficiently Mining Frequent Itemsets"
- By: Karam Gouda & Mohammed J. Zaki, 2005

Item /Tid	A	B	C	D
1	x	x	x	
2			x	x
3	x	x	x	
4	x	x	x	x
5	x			
6		x		x
7			x	



GenMax algorithm

Some Useful Definitions:

- The Combine-Set of an itemset I, is the set of items that can be added to I to create a frequent itemset.
- For example, in the previous example, The combine-set of the itemset {A} is {B, C}.
- The combine-set of the empty itemset is called F1 and is actually the set of frequent itemsets ofsize 1.

GenMax algorithm

- Example:

TID	Items
1	{A,B}
2	{B,C,D}
3	{A,B,C,D}
4	{A,B,D}
5	{A,B,C,D}

minsup=3

GenMax algorithm

```
//invocation : MFI – backtrack(ϕ, F1,0)
MFI – backtrack(Ik, Ck, k)
1.  for each  x ∈ Ck
2.      Ik+1 = Ik ∪ {x}
3.      Pk+1 = {y |: y ∈ Ck  and  y > x}
4.      If  Ik+1 ∪ Pk+1 has a superset in MFI
5.          return
6.      Ck+1 = FI - Combine(Ik+1, Pk+1)
7.      if Ck+1 is empty
8.          if Ik+1 has no superset in MFI
9.              MFI = MFI ∪ Ik+1
10.     else
11.         MFI – backtrack(Ik+1, Ck+1, k +1)
```

```
FI – combine(Ik+1, Pk+1)
1.    C = ϕ
2.    for each y ∈ Pk+1
3.        if Ik+1 ∪ {y} is frequent
4.            C = C ∪ {y}
5.    return C
```

What are the alternatives?

- Is there some other ways of summarizing the frequent itemsets?
- Another popular representation of frequent itemsets is the **closed itemsets** (Pasquier et al., 1999)

Could we do better?

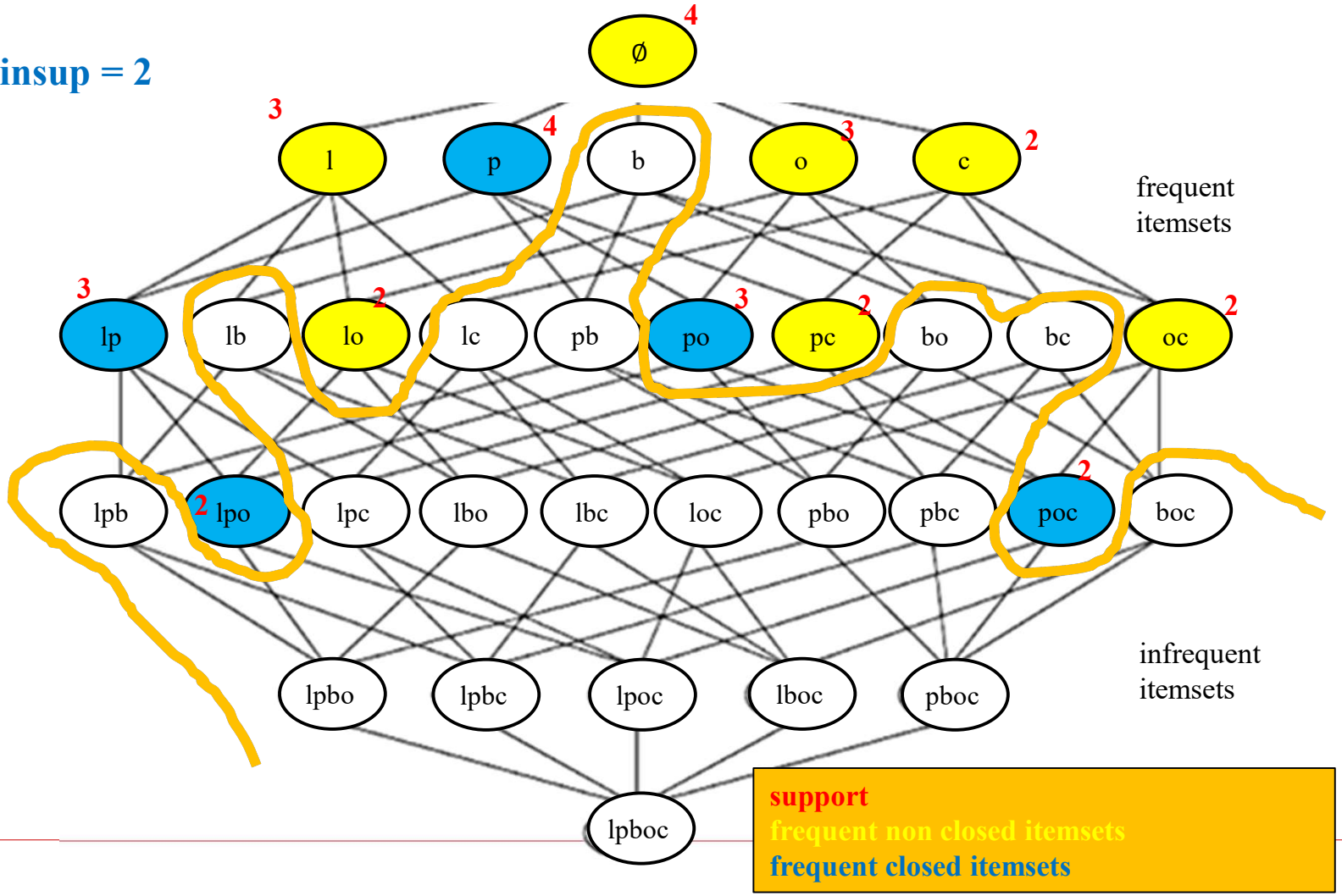
- Is there some other ways of summarizing the frequent itemsets?
- Another popular representation of frequent itemsets is the **closed itemsets** (Pasquier et al., 1999)

Definition: A (frequent) **itemset** X is **closed** if there exists no (frequent) itemset Y such that
$X \subset Y$ and $\sup(X) = \sup(Y)$.

Closed itemsets

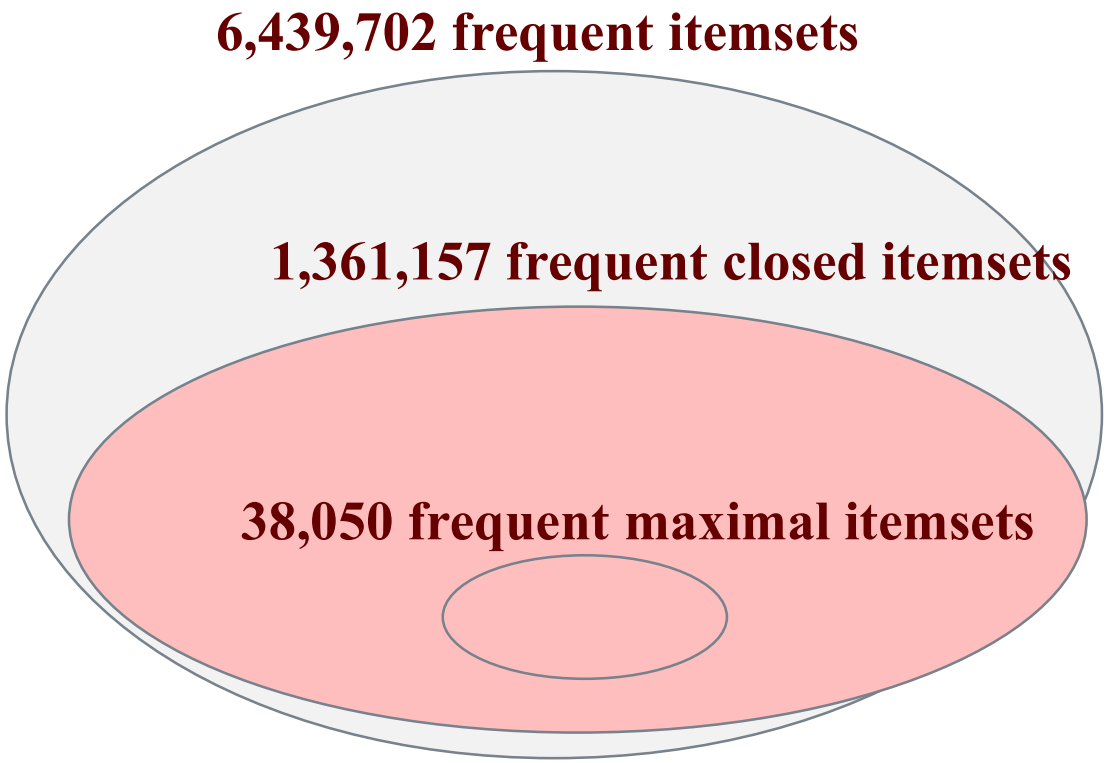
{pasta}	**support = 4**	**closed**
{lemon}	support = 3	
{orange}	support = 3	
{cake}	support = 2	
{pasta, lemon}	**support: 3**	**closed**
{pasta, orange}	**support: 3**	**closed**
{pasta, cake}	support: 2	
{lemon, orange}	support: 2	
{orange, cake}	support: 2	
{pasta, lemon, orange}	**support: 2**	**closed**
{pasta, orange, cake}	**support: 2**	**closed**

minsup =2 The frequent closed itemsets



Closed itemsets are compact

Example: ***Chess*** dataset and minsup =0.4



Recovering all frequent itemsets

Definition: A (frequent) **itemset** X is **closed** if there exists no (frequent) itemset Y such that
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In this example *(minsup = 2)*, there are only five closed itemsets:

{pasta}	**support = 4**	**closed**
{pasta, lemon}	**support: 3**	**closed**
{pasta, orange}	**support: 3**	**closed**
{pasta, lemon, orange}	**support: 2**	**closed**
{pasta, orange, cake}	**support: 2**	**closed**

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{pasta, lemon, orange}	**support: 2**	**closed**
{pasta, orange, cake}	**support: 2**	**closed**

By enumerating all their subsets, we can find all the other frequent itemsets:
{pasta, cake}, {lemon, orange}, {orange, cake},
{lemon,{orange} ,{cake}

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{pasta, lemon}	**support: 3**	**closed**
{pasta, orange}	**support: 3**	**closed**
{pasta, lemon, orange}	**support: 2**	**closed**
{pasta, orange, cake}	**support: 2**	**closed**

By enumerating all their subsets, we can find all the other frequent itemsets:
{pasta, cake}, {lemon, orange}, {orange, cake},
{lemon,{orange} ,{cake}

But how to find the support of these itemsets? →

Recovering all frequent itemsets

Steps to find the support of a frequent itemset X from closed itemsets

1. Find the closure of X.
2. Return the support of the closure of X

Definition: The closure of an **itemset** X is the smallest closed itemset Y such that $X \subseteq Y$. It is denoted as c(X)

Recovering all frequent itemsets

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Example:

{pasta}	**support = 4**	**closed**
{pasta, lemon}	**support: 3**	**closed**
{pasta, orange}	**support: 3**	**closed**
{pasta, lemon, orange}	**support: 2**	**closed**
{pasta, orange, cake}	**support: 2**	**closed**

The closure of {cake} is c({cake}) ???

Recovering all frequent itemsets

Steps to find the support of a frequent itemset X from closed itemsets

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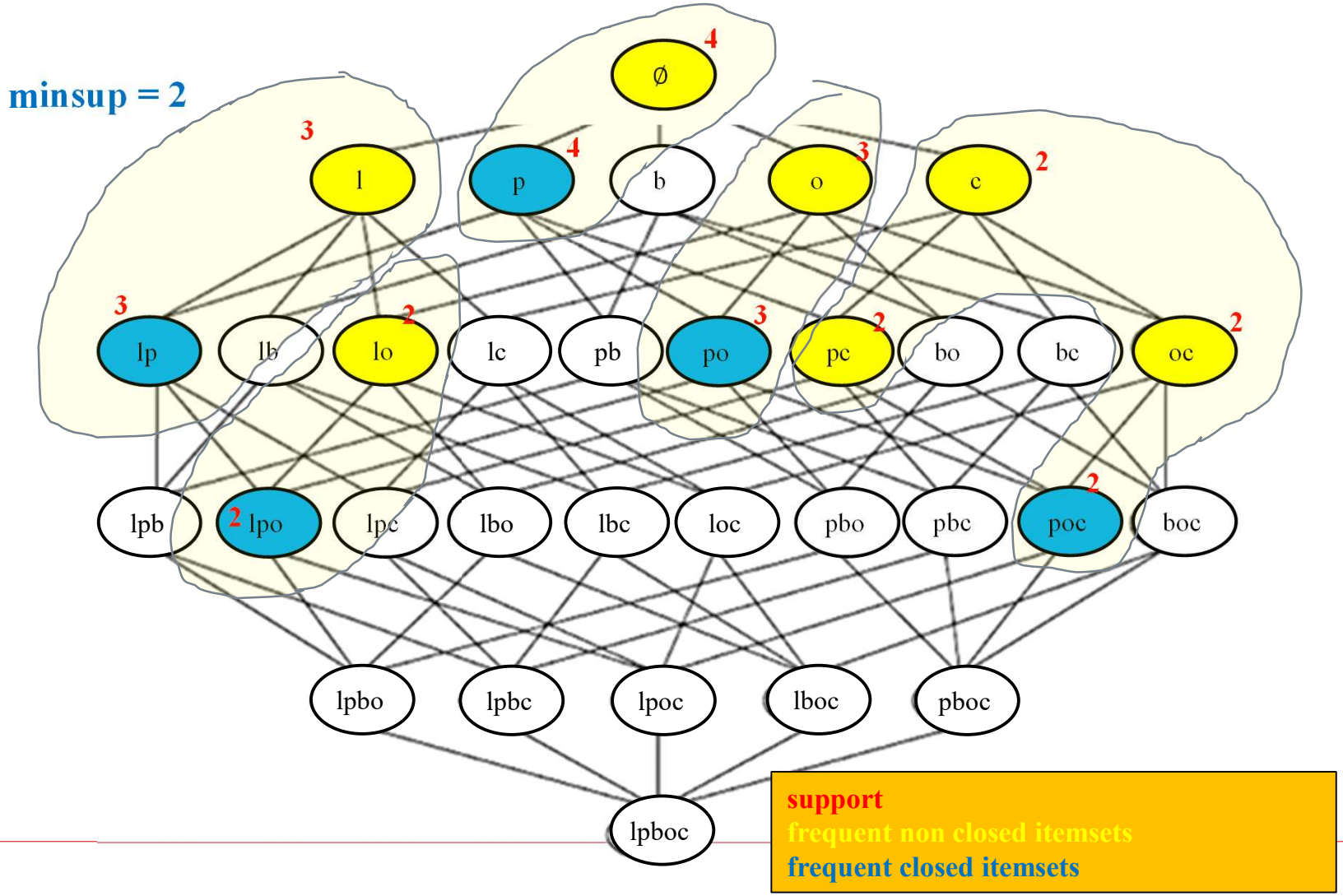
Example:

{pasta}	**support = 4**	**closed**
{pasta, lemon}	**support: 3**	**closed**
{pasta, orange}	**support: 3**	**closed**
{pasta, lemon, orange}	**support: 2**	**closed**
{pasta, orange, cake}	**support: 2**	**closed**

The closure of {cake} is c({cake}) = {pasta,orange,cake}
Thus, the support of {cake} is the same, that is 2.

Itemsets and their closures



Itemsets that have the same closure

- They all have the same support
- They all appear in the same transactions.

Example:

{cake}
{pasta, cake},
{orange, cake},
{pasta, orange, cake}

They all have a support of 2
They all appears in T3 and T4

Trans.	Items
T1	{pasta, lemon, bread, orange}
T2	{pasta, lemon}
T3	{pasta, orange, cake}
T4	{pasta, lemon, orange cake}

Itemsets that have the same closure

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- They all appear in the same transactions.

Example:

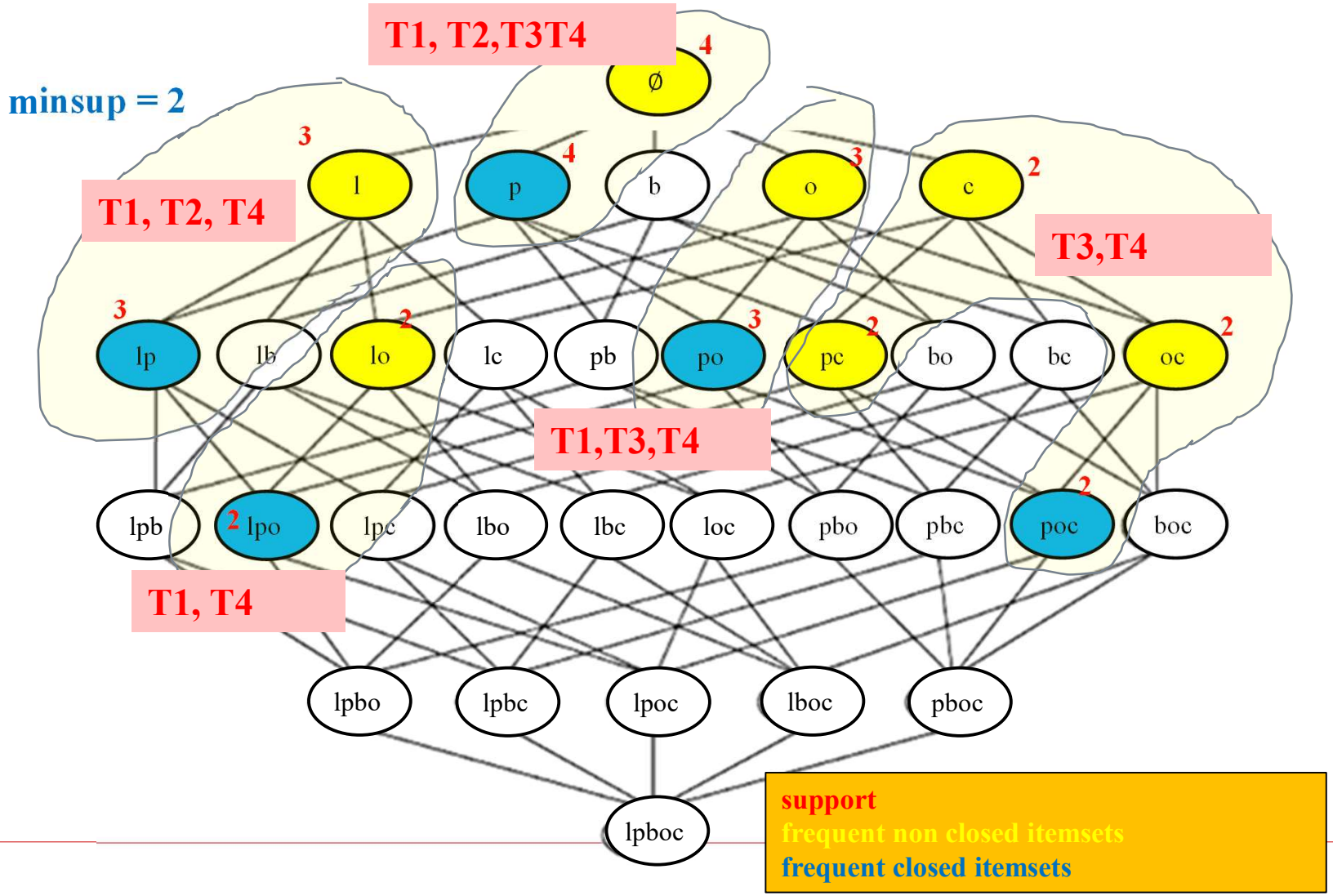
{cake}
{pasta, cake},
{orange, cake},
{pasta, orange, cake}

They all have a support of 2
They all appears in T3 and T4

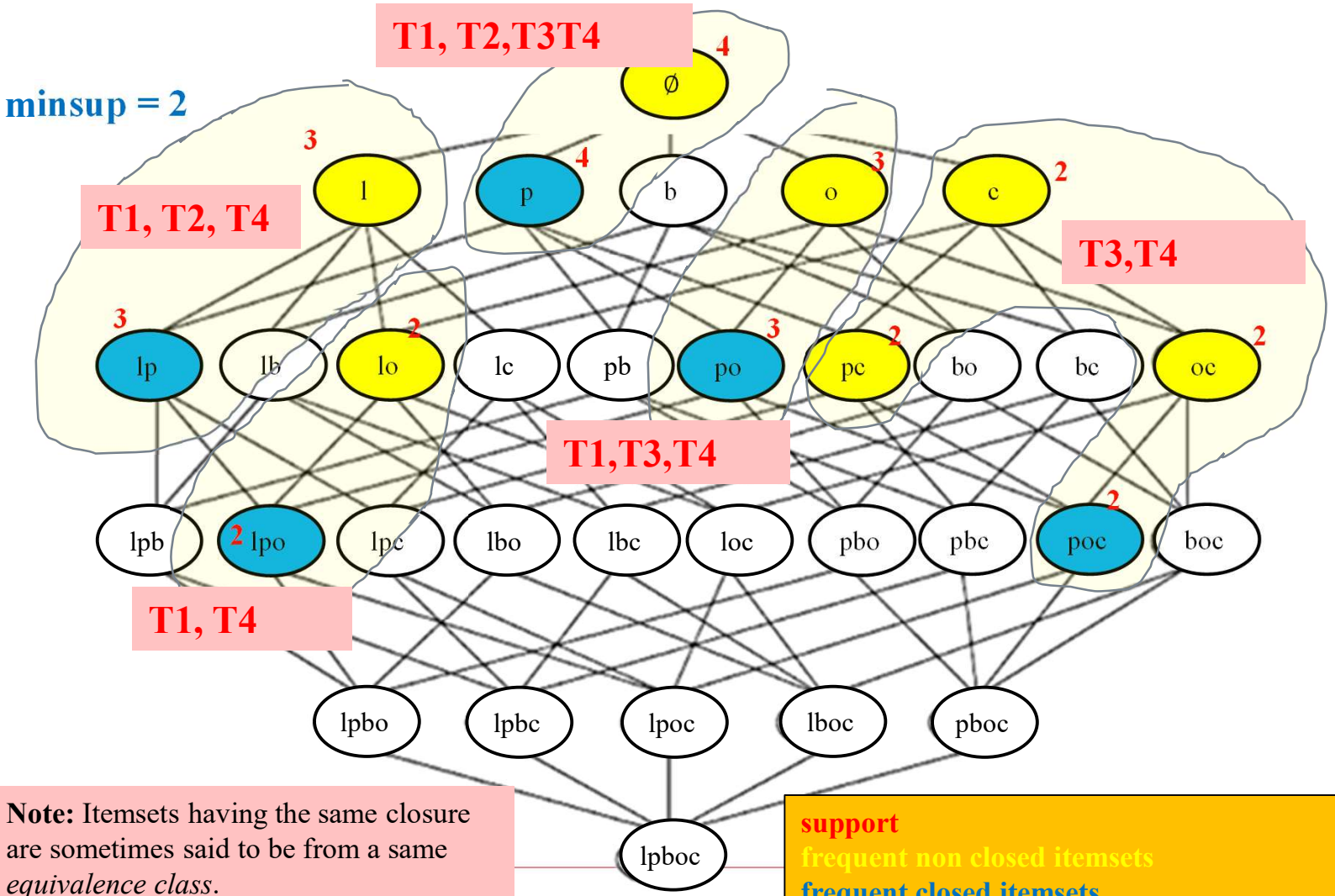
Observation:
The closure is the intersection of transactions T3 and T4

Trans.	Items
T1	{pasta, lemon, bread, orange}
T2	{pasta, lemon}
T3	{pasta, orange, cake}
T4	{pasta, lemon, orange cake}

Itemsets and their closures



Itemsets and their closures



Note: Itemsets having the same closure are sometimes said to be from a same *equivalence class*.

support
frequent non closed itemsets
frequent closed itemsets

How to find the closed itemsets?

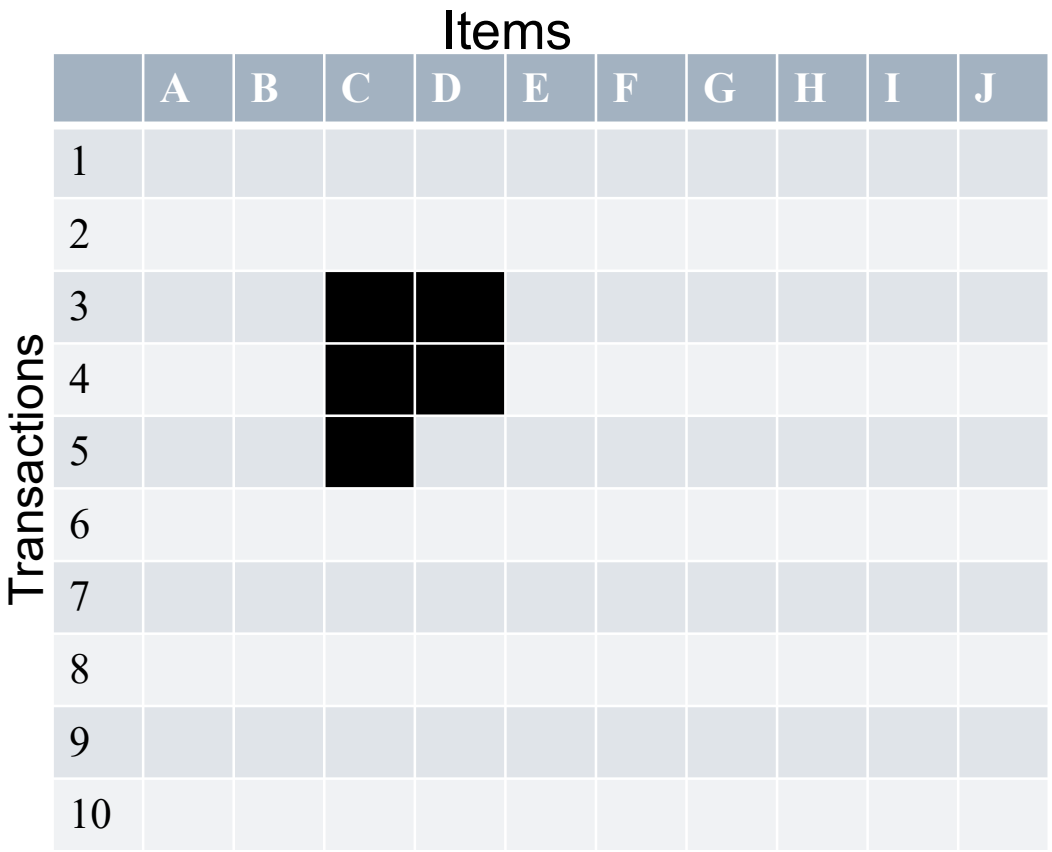
- **Naïve approach**:
  - by post-processing
  - **inefficient!**

TID	Items
1	{A,B}
2	{B,C,D}
3	{A,B,C,D}
4	{A,B,D}
5	{A,B,C,D}

Itemset	Support
{A}	4
{B}	5
{C}	3
{D}	4
{A,B}	4
{A,C}	2
{A,D}	3
{B,C}	3
{B,D}	4
{C,D}	3

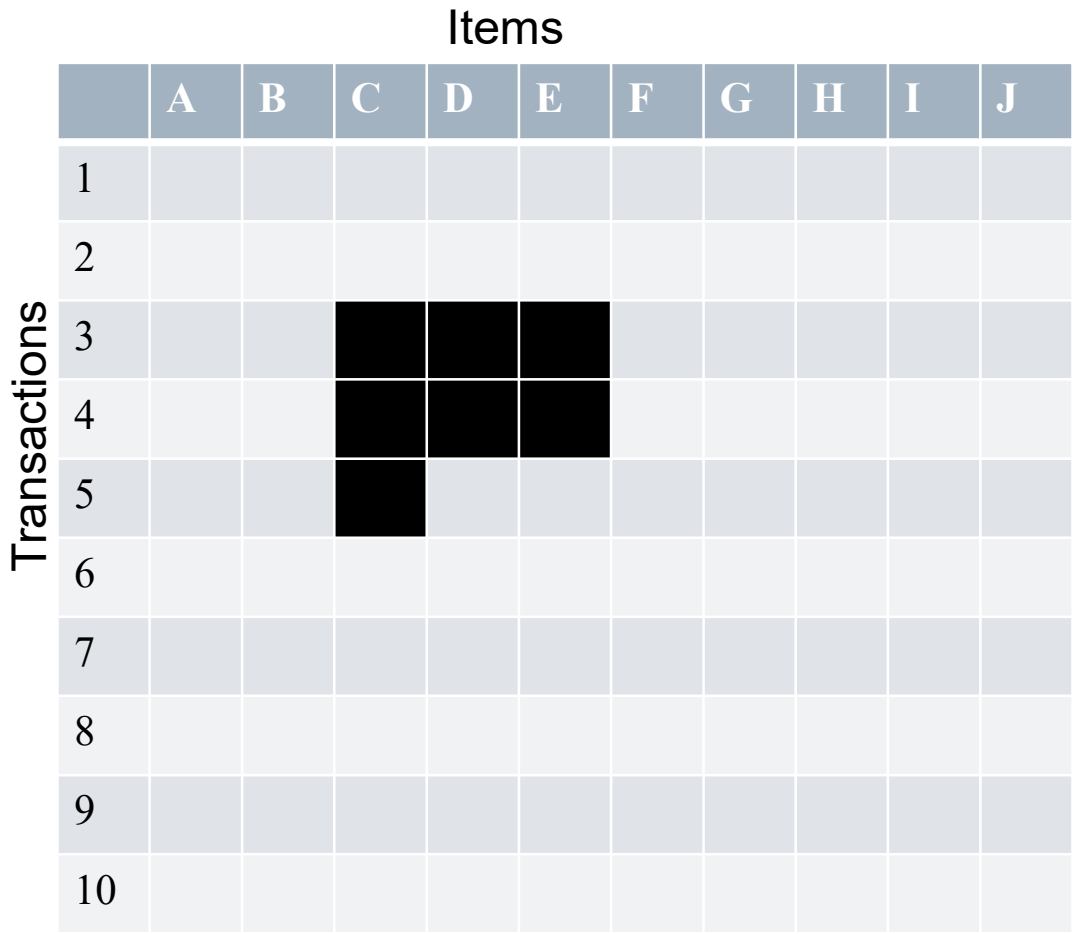
Itemset	Support
{A,B,C}	2
{A,B,D}	3
{A,C,D}	2
{B,C,D}	2
{A,B,C,D}	2

Example 1



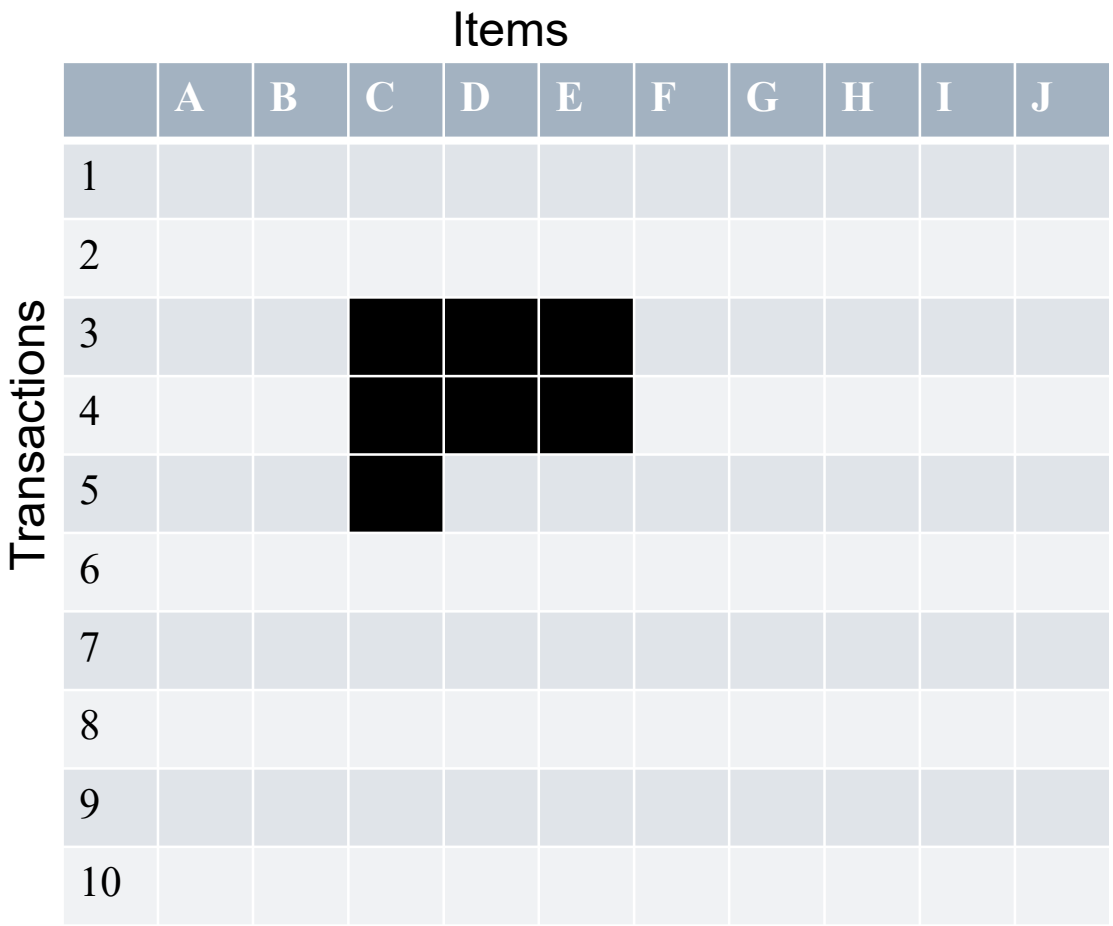
Itemsets	Support (counts)	Closed itemsets
{C}	**3**	✓
{D}	2	
{C,D}	**2**	✓

Example 2



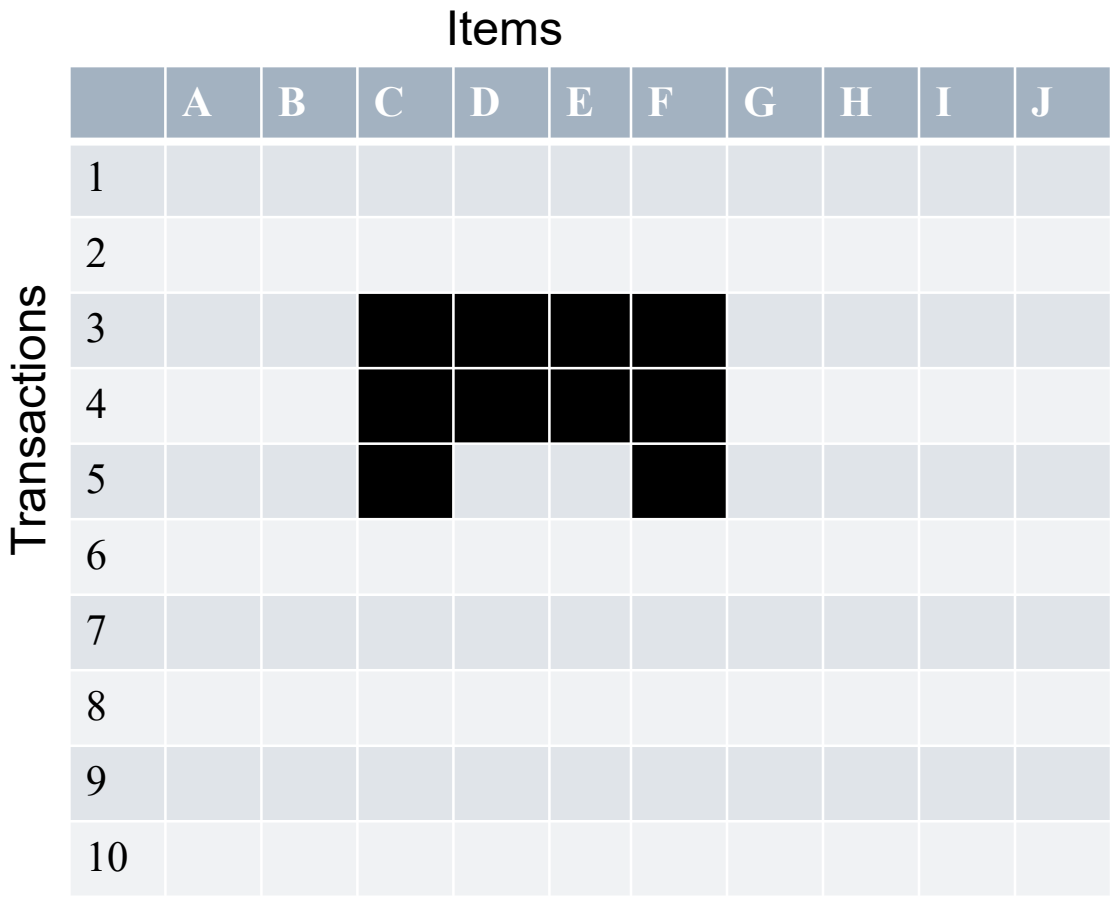
Itemsets	Support (counts)	Closed itemsets
{C}	3	
{D}	2	
{E}	2	
{C,D}	2	
{C,E}	2	
{D,E}	2	
{C,D,E}	2	

Example 2



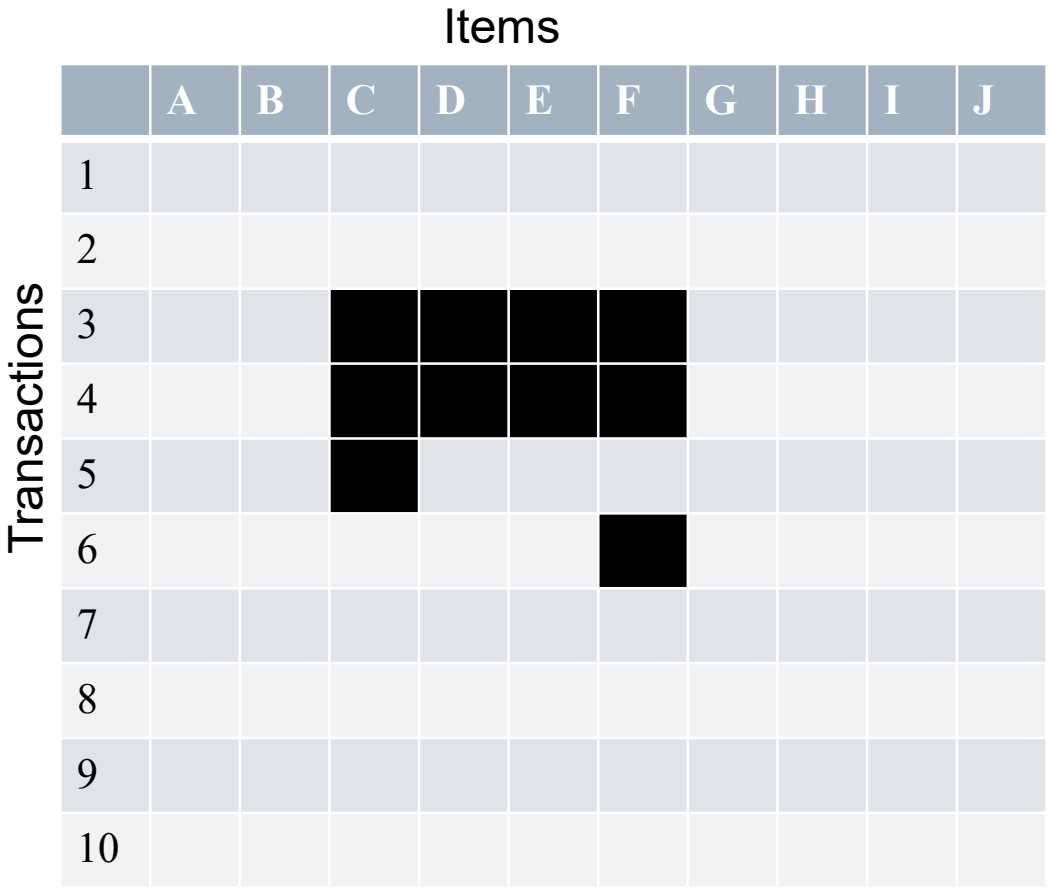
Itemsets	Support (counts)	Closed itemsets
{C}	**3**	✓
{D}	2	
{E}	2	
{C,D}	2	
{C,E}	2	
{D,E}	2	
{C,D,E}	**2**	✓

Example 3



Closed itemsets: {C,D,E,F}, {C,F}

Example 4



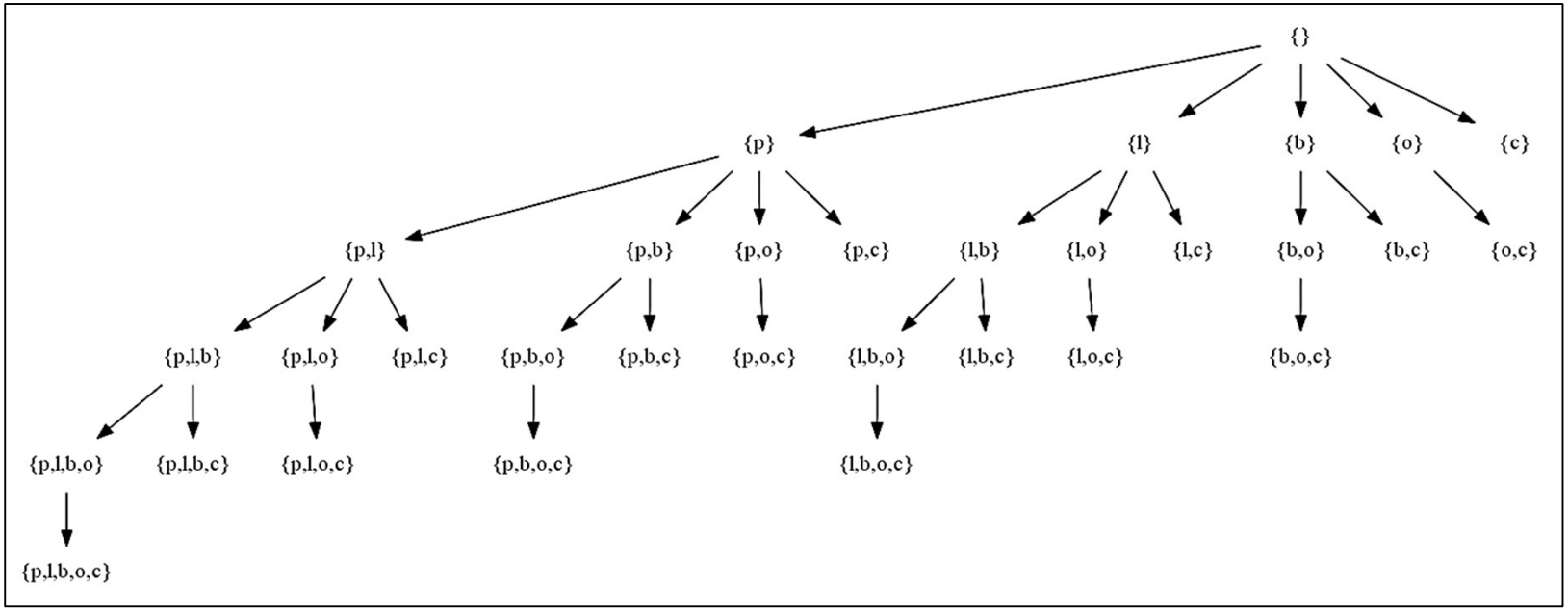
Closed itemsets: {C,D,E,F}, {C}, {F}

How to find the closed itemsets?

- **Efficient algorithms:**
  - Charm, DCI_Closed,
    FPClose, Closet, LCM, etc.
  - Find frequent itemsets by avoiding generating all frequent itemsets.
  - Several strategies and properties.

Charm (Zaki, 2001)

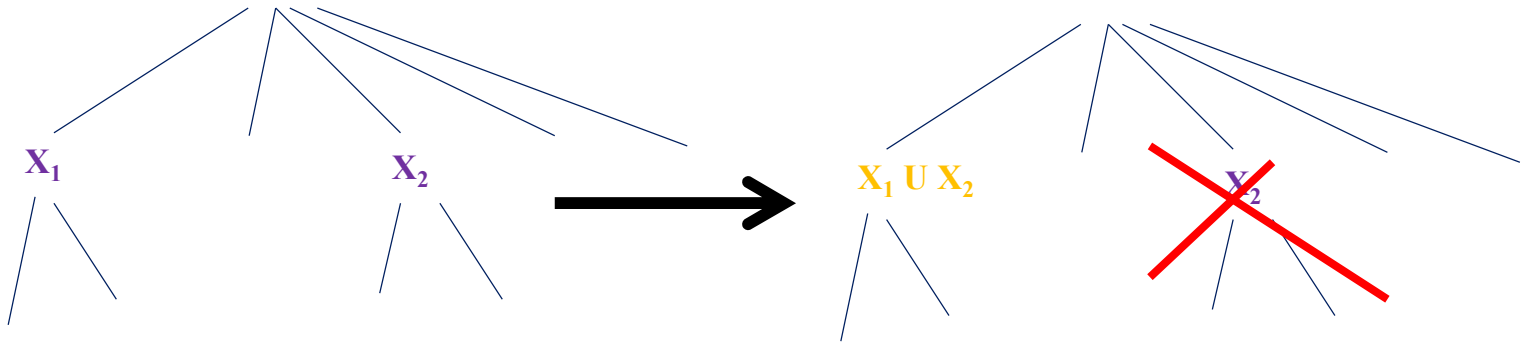
- Based on ECLAT
- Depth-first search



- **Four cases must be checked** to avoid generating all frequent itemsets→

Four cases

Suppose that the algorithm is considering combining an itemset X_1 with an itemset X_2 such that $X_2 \succ X_1$.



Property 1

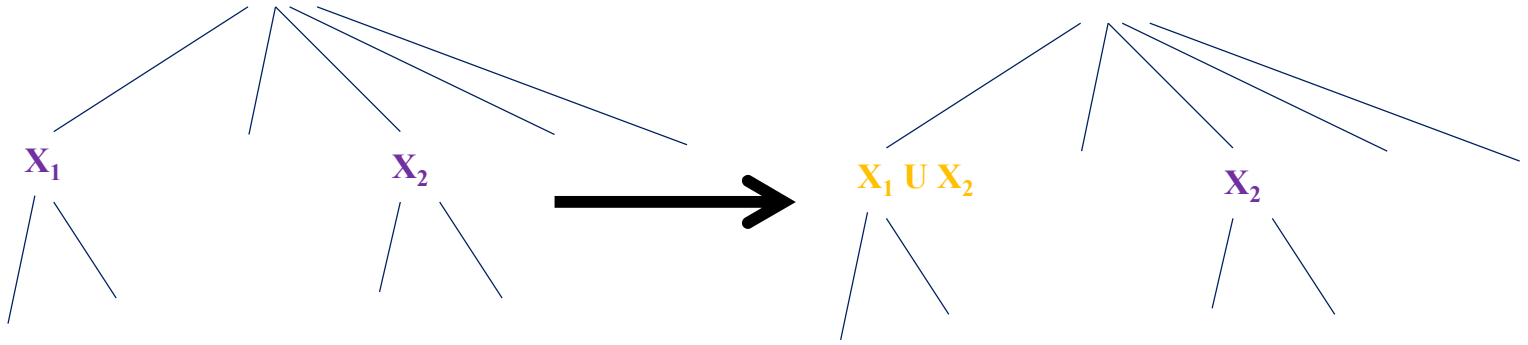
IF $t(X_1) = t(X_2)$, **THEN** $t(X_1) = t(X_2) = t(X_1 \cap X_2)$.

- Thus, we can replace all occurrences of X_1 by $X_1 \cup X_2$.
- We do not need to consider X_2.

Notation: t(X) denotes the set of transactions containing an itemset X

Four cases

Suppose that the algorithm is considering combining an itemset X_1 with an itemset X_2 such that $X_2 \succ X_1$.



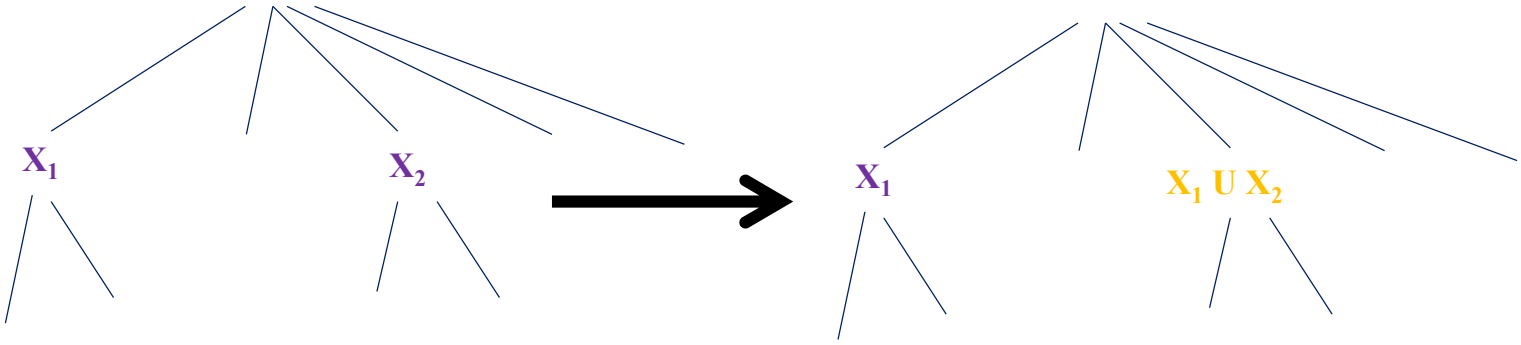
Property 2

IF $t(X_1) \subset t(X_2)$, **THEN** $t(X_1) = t(X_1 \cap X_2) \neq t(X_2)$.

- Thus, we can replace all occurrences of X_1 by $X_1 \cup X_2$.
- But we must keep X_2.

Four cases

Suppose that the algorithm is considering combining an itemset X_1 with an itemset X_2 such that $X_2 \succ X_1$.



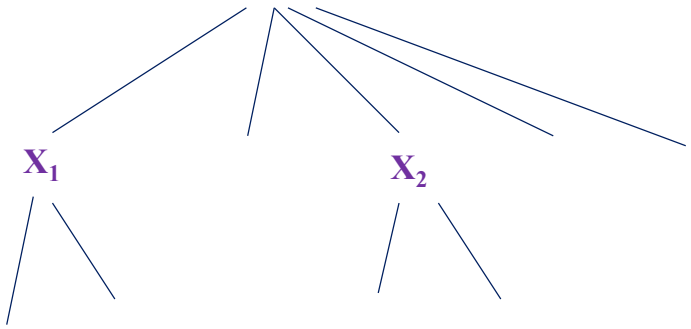
Property 3

IF $t(X_1) \supset t(X_2)$, **THEN** $t(X_2) = t(X_1 \cap X_2) \neq t(X_1)$.

- Thus, we can replace all occurrences of X_2 by $X_1 \cup X_2$.
- But we must keep X_1.

Four cases

Suppose that the algorithm is considering combining an itemset X_1 with an itemset X_2 such that $X_2 \succ X_1$.



Property 4

IF $t(X_1) \neq t(X_2)$, **THEN** $t(X_2) \neq t(X_1 \cap X_2) \neq t(X_1)$.

- We cannot remove anything.

Charm

- These four properties allows to
  - eliminate several frequent itemsets
  - thus, reduce the search space
- However, the algorithm can still generate several itemsets that are non closed.
- We must add a **closure checking test**.

Closure checking

- For each **frequent itemset X found**, we compare it to the other **frequent closed itemsets** found until now.
- If **X** is contained in an itemset that has been already found, we do not keep **X.**
- Otherwise, **X** is closed. It is added to the set of **closed itemsets**.
- How to implement this test?
  - store closed itemsets in a hash table
  - hash function is the list of transaction ids t(.).

Pseudocode of Charm

```
CHARM (δ ⊆ I × T, minsup):
1.  Nodes = {I_j × t(I_j) : I_j ∈ I ∧ |t(I_j)| ≥ minsup}
2.  CHARM-EXTEND (Nodes, C)

CHARM-EXTEND (Nodes, C):
3.  for each X_i × t(X_i) in Nodes
4.      NewN = ∅ and X = X_i
5.      for each X_j × t(X_j) in Nodes, with f(j) > f(i)
6.          X = X ∪ X_j and Y = t(X_i) ∩ t  j > i
7.          CHARM-PROPERTY(Nodes, NewN)
8.      if NewN ≠ ∅ then CHARM-EXTEND (NewN)
9.      C = C ∪ X //if X is not subsumed

CHARM-PROPERTY (Nodes, NewN):
10. if (|Y| ≥ minsup) then
11.     if t(X_i) = t(X_j) then //Property 1
12.         Remove X_j from Nodes
13.         Replace all X_i with X
14.     else if t(X_i) ⊂ t(X_j) then //Property 2
15.         Replace all X_i with X
16.     else if t(X_i) ⊃ t(X_j) then //Property 3
17.         Remove X_j from Nodes
18.         Add X × Y to NewN
19.     else if t(X_i) ≠ t(X_j) then //Property 4
20.         Add X × Y to NewN
```

Nodes the set of frequent itemsets of size 1
I *is the set of items* ***T*** *is the set of transactions*
C *is the set of all closed itemsets found*
X x t(X) *denotes an* ***X*** *and its list of transaction* ***t(X)***

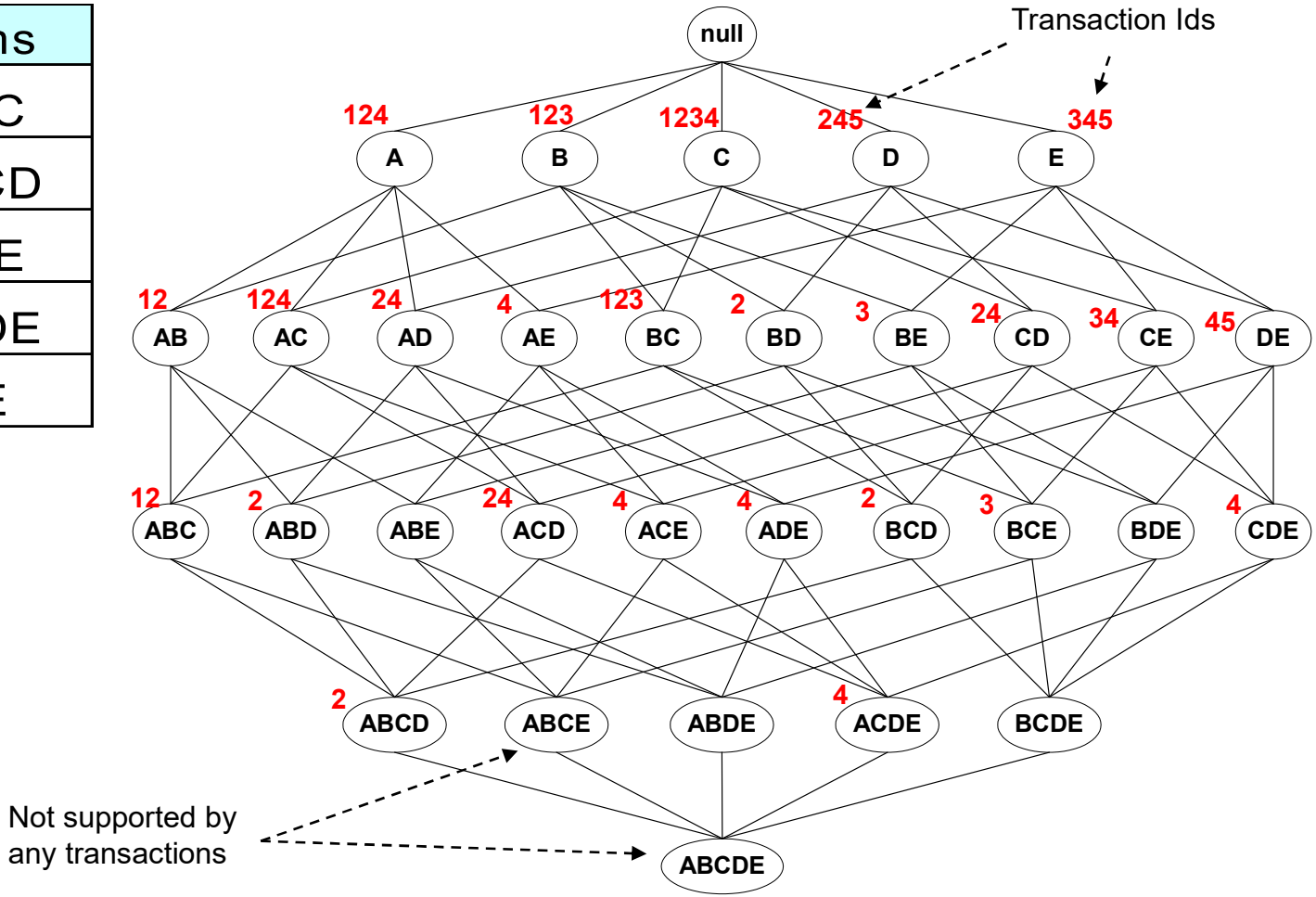
Maximal vs Closed Itemsets

- **Closed frequent itemsets**
  - Can recover all frequent itemsets
  - Can recover the support
- **Maximal frequent itemsets**
  - Can recover all frequent itemsets
  - Cannot recover the support
  - A subset of closed itemsets, sometimes much smaller.

Is there other compact representations of frequent itemsets? Yes →

Maximal vs Closed Itemsets

TID	Items
1	ABC
2	ABCD
3	BCE
4	ACDE
5	DE



Maximal Frequent vs Closed Frequent Itemsets

TID	Items
1	ABC
2	ABCD
3	BCE
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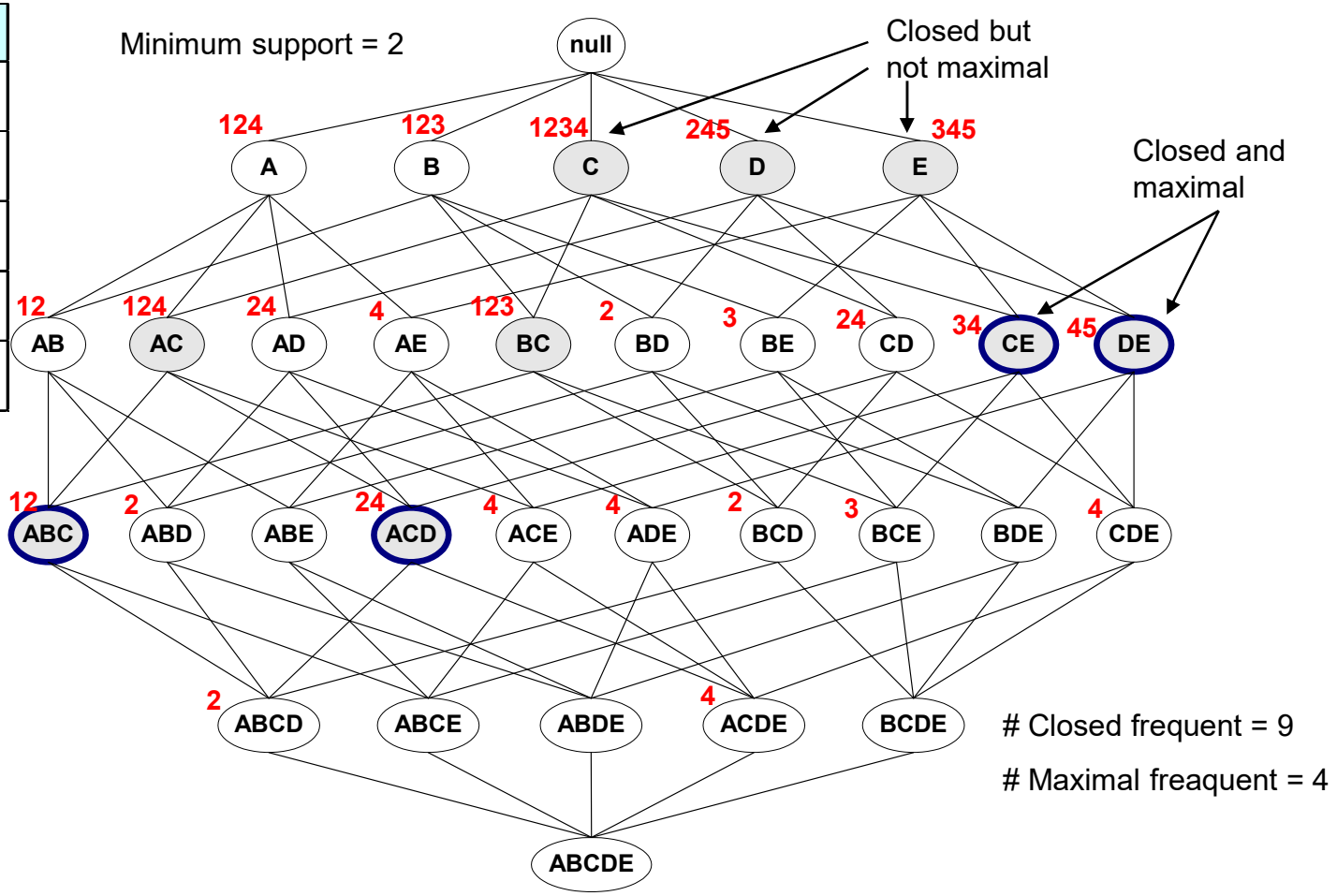
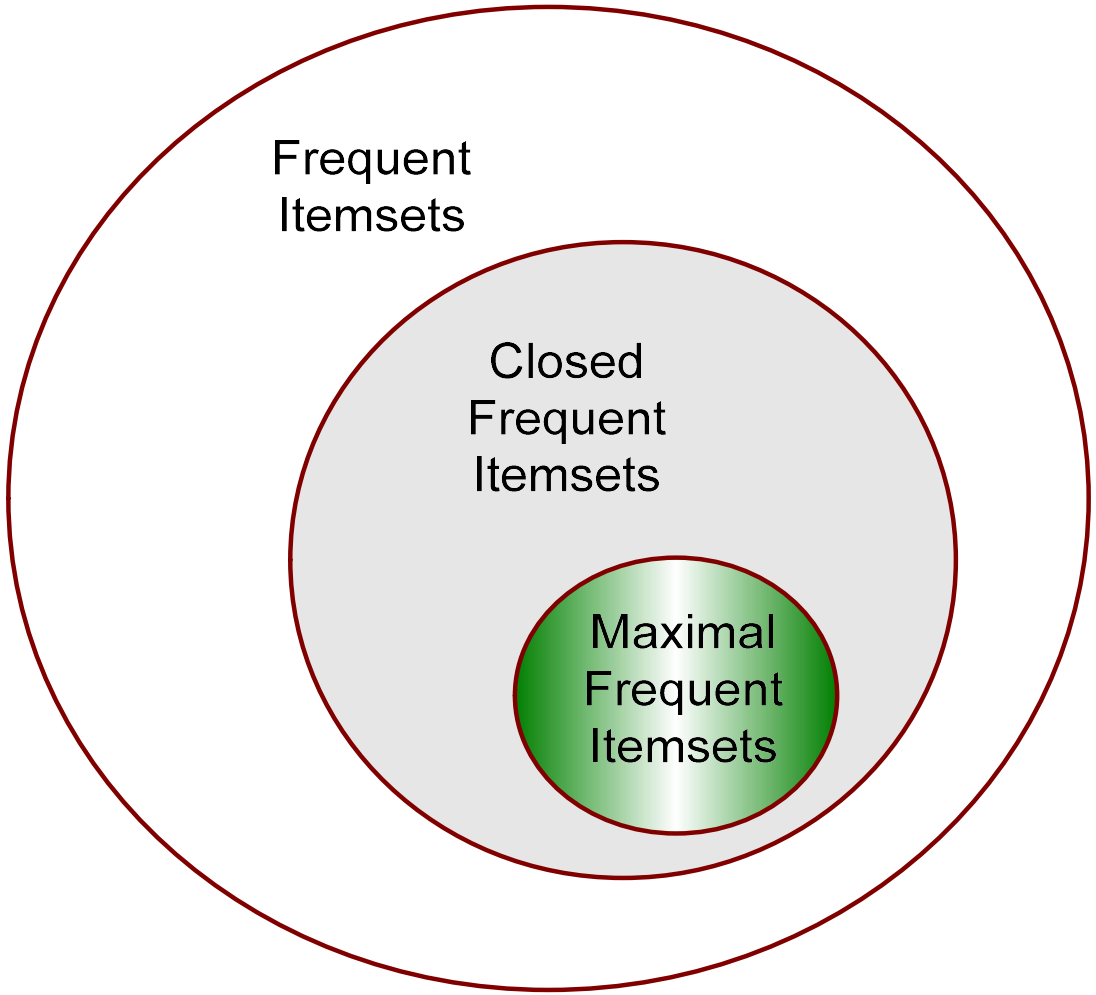


Maximal vs Closed Itemsets



The generator itemsets (or key itemsets)

Definition: A (frequent) **itemset** X is a **generator** if there exists no (frequent) itemset Y such that $Y \subset X$ and $\sup(X) = \sup(Y)$.

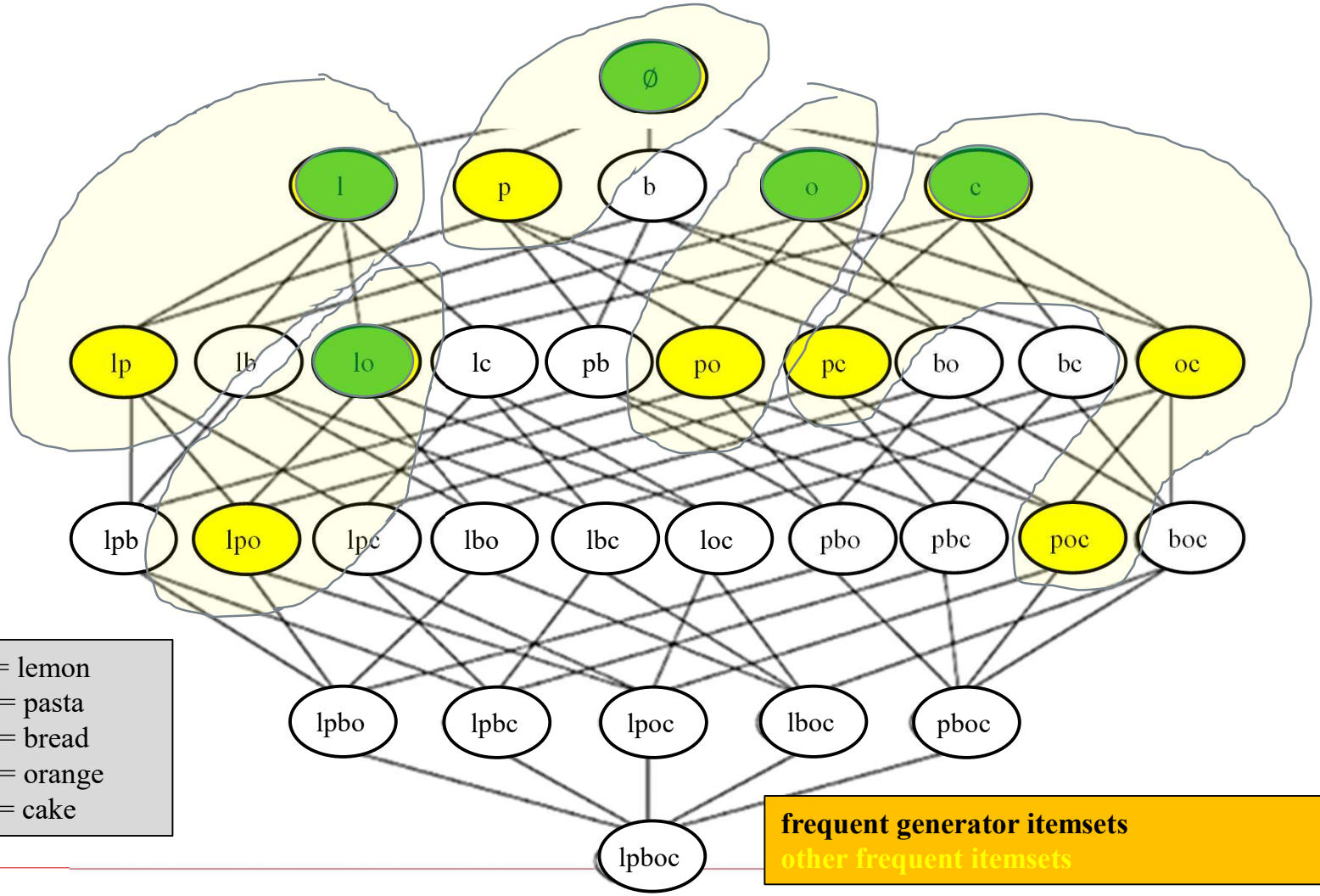
The generator itemsets (or key itemsets)

Definition: A (frequent) **itemset** X is a **generator** if there exists no (frequent) itemset Y such that $Y \subset X$ and $\sup(X) = \sup(Y)$.

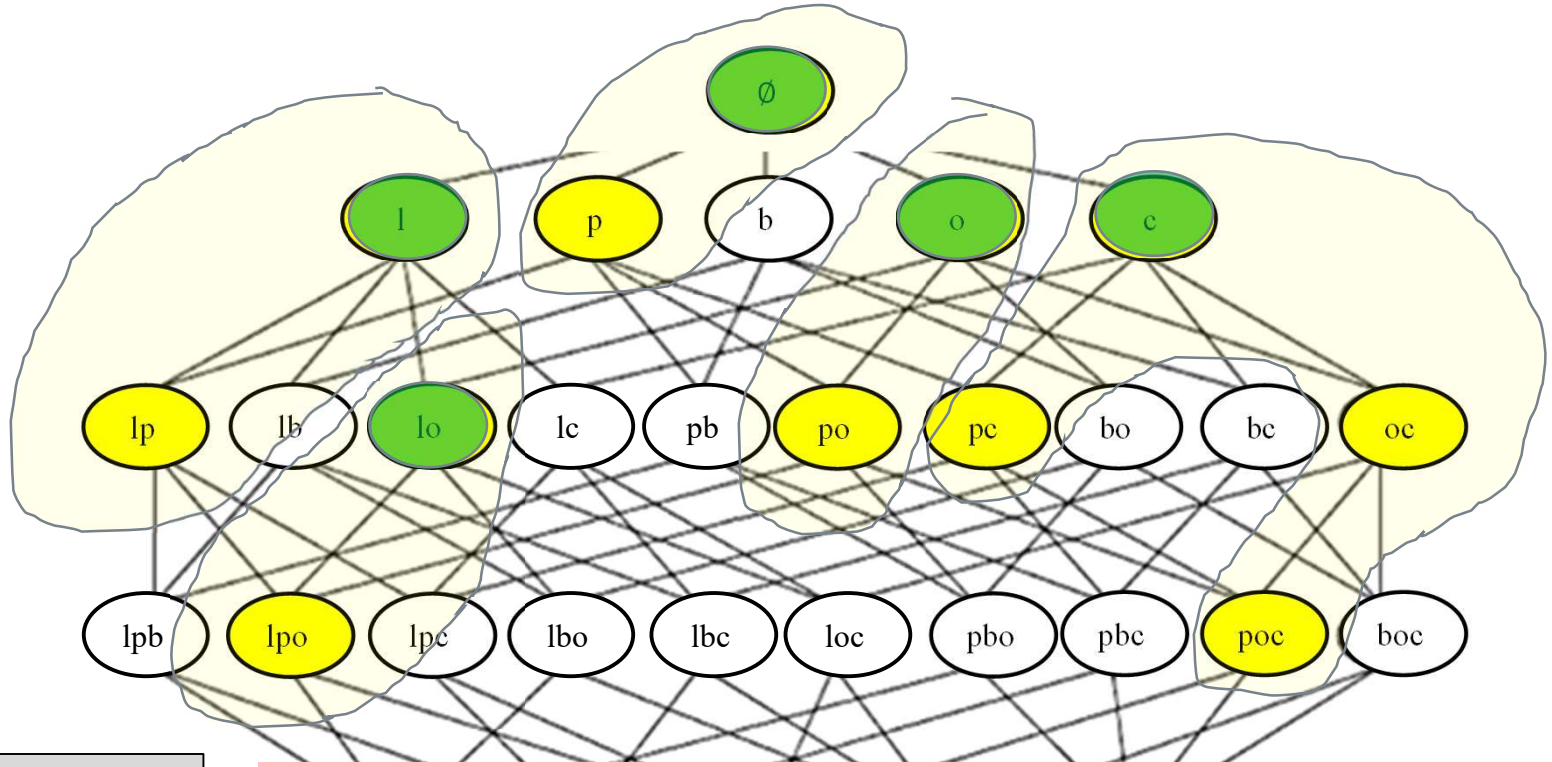
In the example *(minsup = 2)*, there are five **frequent generator itemsets**:

{}	support: 4
{lemon}	support: 3
{orange}	support: 3
{cake}	support: 2
{lemon,orange}	support: 2

minsup =2 The frequent generator itemsets



minsup =2 The frequent generator itemsets



l = lemon
p = pasta
b = bread
0 = orange
c = cake

Some observations:

- The **empty set** can be a generator
- While each equivalence class has always one closed itemset, it could have more than one generator (not in this example)
- In some cases, a closed itemset can be a generator (not in this example)

Why generators are interesting?

- They are the smallest sets of items that describe a set of transactions.
  e.g. the smallest sets of products common to a group of customers.
- Some studies have shown shown that generator patterns can provide better accuracy for classification than maximal or closed itemsets.

Why generators are interesting?

- Also, if we find **generator** and **closed** itemsets, we can use them to generate compact representations of association rules of the form:

generator → closed itemset - generator

{cake} → {pasta, orange} support: 2
confidence: 100%

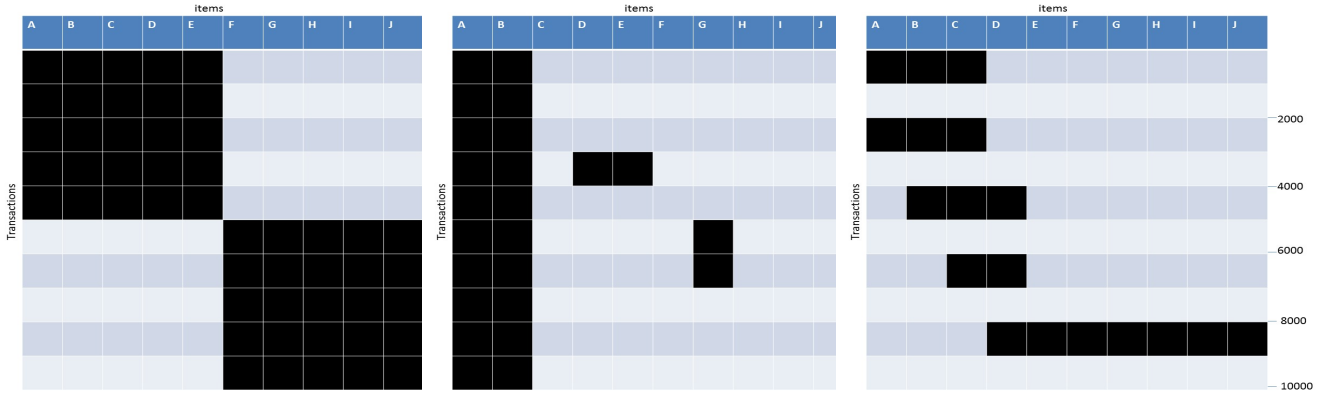
generator: {cake}
closed: {pasta,orange,cake}

How to find the generator itemsets?

- **Naïve approach**:
  - by post-processing
  - **inefficient!**
- **Efficient algorithms:**
  - Pascal, DefMe, etc.
  - Find generator itemsets by avoiding generating all frequent itemsets.
  - **Key search space pruning property:**
    An itemset can only be a generator if all its subsets are also generators.

Example question

- Given the following transaction data sets (dark cells indicate presence of an item in a transaction) and a support threshold of 20%, answer the following questions



DataSet: A Data Set: B Data Set: C

a. What is the number of frequent itemsets for each dataset? Which dataset will produce the most number of frequent itemsets?
b. Which dataset will produce the longest frequent itemset?
c. Which dataset will produce frequent itemsets with highest maximum support?
d. Which dataset will produce frequent itemsets containing items with widely varying support levels (i.e., itemsets containing items with mixed support, ranging from 20% to more than 70%)?
e. What is the number of maximal frequent itemsets for each dataset? Which dataset will produce the most number of maximal frequent itemsets?
f. What is the number of closed frequent itemsets for each dataset? Which dataset will produce the most number of closed frequent itemsets?